

Comparative Analysis of Deep Learning Models for Detecting Small Underwater Objects in Deep-Sea Environments



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Abstract

This thesis presents a comparative evaluation study of eight modern deep learning-based object detection models for underwater small-object detection. Underwater object detection is a challenging computer vision task because underwater environments contain severe visual degradations such as low illumination, scattering effects, color distortion, turbidity, motion blur, and cluttered backgrounds. These environmental conditions significantly affect image quality and make accurate localization of underwater objects difficult, particularly for small objects with limited visual features.

The study evaluates eight state-of-the-art object detection architectures, including YOLOv8n, YOLOv10n, RT-DETR-L, Faster R-CNN, RetinaNet, SSDlite320-MobileNetV3, Mask R-CNN ResNet50-FPN, and DETR ResNet-50. The experiments were conducted using an underwater object detection dataset containing six object categories such as propeller, pipe_type2, red_fin, net, qr_codes, and pipe. All models were trained and evaluated under identical experimental conditions to ensure fair comparative analysis.

The performance of the models was evaluated using multiple accuracy-based and efficiency-based metrics including $mAP@0.5$, $mAP@0.5:0.95$, precision, recall, detection accuracy, parameter count, training time, inference time, and frames per second (FPS). The results demonstrated that YOLOv8n achieved the best overall balance between detection accuracy and computational efficiency, achieving an $mAP@0.5$ score of 98.46% and FPS of 113.67. RT-DETR-L achieved the best strict localization performance under $mAP@0.5:0.95$ evaluation, while Faster R-CNN achieved the highest recall performance. SSDlite320-MobileNetV3 demonstrated strong computational efficiency but lower detection accuracy. DETR ResNet-50 produced the weakest overall performance under the experimental settings used in this study.

The findings of this thesis contribute a comprehensive benchmark comparison of modern deep learning-based object detection architectures for underwater small-object detection. The results may support future research and practical deployment of underwater computer vision systems in applications such as underwater robotics, marine monitoring, autonomous underwater vehicles, and underwater inspection systems.

Preface

This thesis was prepared as part of the Master's Degree Programme at Åbo Akademi University. The work presented in this report reflects the knowledge and experience gained during my studies and research activities in the field of computer science and parallel computing.

I would like to express my sincere gratitude to my supervisor, Professor Johan Lilius, for his valuable guidance, support, and constructive feedback throughout this work. His expertise and encouragement greatly contributed to the successful completion of this thesis.

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Md Anzir Hossain Rafath

Declaration of AI Tool Usage

During the preparation of this thesis, limited artificial intelligence (AI) based tools were used in this study as a supportive tool during different stages of the research and writing process. These tools were mainly used to help with brainstorming research ideas, improving academic writing structure, grammar correction, and getting coding-related suggestions during the implementation phase.

Furthermore, GitHub Copilot tool was mainly used for coding assistance purpose such as debugging support, implementation suggestions, and Python programming during the experimental development process.

Additionally, ChatGPT (OpenAI) tool was used to assist background exploration of the existing research area, and coding-related guidance for deep learning model implementation and getting models evaluation process suggestions of this research.

Another AI tool Grammarly was mainly used for grammar checking, spelling correction, and final language refinement during thesis editing

Overall, all of the experimental implementations, model training procedures, description of results, table and figures visualizations, interpretations, and final thesis content were reviewed, verified, modified, and validated by the author before inclusion in the thesis.

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List of Abbreviations

AI	Artificial Intelligence
AP	Average Precision
AUV	Autonomous Underwater Vehicle
CNN	Convolutional Neural Network
COCO	Common Objects in Context
CUDA	Compute Unified Device Architecture
CNN	Convolutional Neural Network
CSP	Cross Stage Partial
DETR	Detection Transformer
DL	Deep Learning
FPN	Feature Pyramid Network
FPS	Frames Per Second
GPU	Graphics Processing Unit
IMMS	Intelligent Marine Monitoring Systems
IoU	Intersection over Union
mAP	Mean Average Precision
ML	Machine Learning
PyTorch	Python-based Deep Learning Framework
QR	Quick Response
R-CNN	Region-based Convolutional Neural Network
RGB	Red Green Blue
ROV	Remotely Operated Vehicle
RPN	Region Proposal Network
RT-DETR	Real-Time Detection Transformer
SSD	Single Shot MultiBox Detector

TP	True Positive
FP	False Positive
FN	False Negative
YOLO	You Only Look Once

Chapter 1 Introduction

1.1 Background of the Study

In the field of underwater applications, underwater environments are important in marine research for environmental monitoring, inspection of offshore infrastructure, underwater robotics, defense and ocean resource exploration. With the development of autonomous underwater vehicles (AUVs), remotely operated vehicles (ROVs), and intelligent marine monitoring systems have achieved significant reliability for underwater object detection technologies (Jian et al., 2024). When it comes to completing some complex tasks such as underwater navigation, pipeline inspection, observation of marine biodiversity, debris monitoring and even underwater rescue operations, it is essential to detect underwater objects accurately.

Although underwater object detection has gained significant progress in the recent years with the advancement of computer vision, it is still one of the most challenging tasks in underwater imaging environment because of the complex characteristics of the underwater object environment. Because water does not absorb and scatter light as much as air does, the contrast, color and visibility are low, and also there are motion blurs and turbidity. The visibility of objects and image clarity is greatly affected by light scattering and attenuation which occur underwater (Jaffe, 2002). So, these visual impairments make underwater object detection significantly more difficult than the traditional object detection in terrestrial images. Underwater environment also faces difficulties due to the complexity of backscattering backgrounds, varying illumination, and particles suspended in the water which further affect the detection performance (Jian et al., 2024).

The challenge in deep-sea environment becomes even greater when detecting small underwater objects. Small objects occupy only a small number of pixels within an image and are often partially obscured or almost indistinguishable from background objects. The problem of small-object detection is one of the most challenging problems in computer vision as small objects are characterized with limited object features which decreases localization accuracy and classification reliability (Kisantal et al., 2019). Furthermore, this problem increases more when the conditions are underwater because of the environmental distortion and the lack of resolution of the objects.

Traditional image processing and handcrafted feature-based methods often struggle to handle the complexity of underwater environments. As a result, deep learning-based object detection methods have become increasingly popular because of their ability to automatically learn hierarchical visual features directly from data. Deep learning techniques enable end-to-end feature learning and significantly improve performance in visual recognition tasks (LeCun et al., 2015).

Underwater environments can be complex, and traditional image processing and handcrafted feature-based methods can often struggle with this complexity for detection objects. Therefore, the use of deep learning-based object detection methods has gained popularity due to its ability to learn visual features at multiple levels automatically from the data. Additionally, another ability to end-to-end feature learning is possible with deep learning methods, and the performance in visual

recognition tasks can be significantly enhanced with the help of deep learning Methods (LeCun et al., 2015).

Recent advancements in deep learning have introduced highly effective object detection models such as YOLO, Faster R-CNN, RetinaNet, DETR, and RT-DETR. These architectures represent different detection paradigms and trade-offs between speed and accuracy, including real-time one-stage detectors and accuracy-focused two-stage or transformer-based models (Redmon et al., 2016; Ren et al., 2015; Carion et al., 2020). Some models are optimized for real-time detection, while others prioritize high detection precision. However, the performance of these architectures under underwater conditions, particularly for small-object detection, remains insufficiently explored. Recent surveys confirm that underwater object detection is still an open research problem due to dataset limitations and environmental complexity (Chen et al., 2024).

Another point is there has been significant progress in deep learning that has introduced highly effective state-of-the-art models for object detection like YOLO, Faster R-CNN, RetinaNet, DETR and RT-DETR. These architectures are based on different detection terms and balance between speed and accuracy such as real-time one-stage detectors, accuracy-focused two-stage or transformer-based models (Ren et al., 2015; Redmon et al., 2016; Carion et al., 2020). There are some models which are optimized for real-time detection, and some models are optimized to have high detection precision. Their performances in underwater environments have not been sufficiently investigated, especially for the detection of small objects. However, underwater object detection remains a challenging research question, as evidenced by recent surveys (Chen et al., 2024) where limited datasets and environmental complexities are identified as challenges.

This thesis study focuses on the comparative analysis of eight modern deep learning-based object detection models for underwater small object detection in deep-sea environments. For this research, the selected models were using for detection underwater objects detection by using different models such as YOLOv8n, YOLOv10n, RT-DETR-L, Faster R-CNN, RetinaNet, SSDlite320-MobileNetV3, Mask R-CNN ResNet50-FPN, and DETR ResNet-50. These models represent advanced developments in object detection, including lightweight real-time architectures and transformer-based detectors (Zhao et al., 2024). In addition, these models are evaluated on the same dataset with many performance measures such as metrics including mAP@0.5, mAP@0.5:0.95, precision, recall, detection accuracy, inference time, and frames per second (FPS). By comparing this analysis, the research aims to find the best deep learning architecture for small object detection in underwater settings.

1.2 Problem Statement

The imaging environment in the water has many serious imaging distortion phenomena, including low illumination, scattering phenomenon, suspended particles and background complexity, which are very serious factors that limit the performance of the underwater object detection system. The

author highlighted that these environmental factors have a significant impact on image quality and detection performance (Jian et al., 2024). Moreover, these problems make the use of traditional object detection methods less effective and find accurate localization of underwater objects challenging.

Although many advanced deep learning-based object detection architectures have illustrated strong performance in land-based datasets, their effectiveness in underwater environments remains uncertain how well such models will perform underwater. Different deep learning models highlighted different trade-offs between accuracy of detection, computational complexity, and inference speed. For example, lightweight models may have achieved real-time better performance but gained poor performance in finding localization accuracy, while large transformer-based models may provide stronger contextual understanding but there is more cost of computational efficiency.

Furthermore, very few comparative studies are available that evaluate the multiple modern object detection architectures under the same underwater experimental setting that use the same consistent evaluation metrics. As a result, this has led to a lack of knowledge about suitable deep learning models for underwater small-object detection tasks.

Moreover, there is a need for comparative evaluation analysis of modern deep learning-based object detection models in underwater environments to identify their strengths, limitations, and feasibility in real life applications.

1.3 Research Gap

Recent studies explained that comprehensive benchmarking across modern architectures in underwater conditions is still limited (Chen et al., 2024). Previous studies showed that the research on underwater object detection mainly focuses on individual detection models or any specific underwater enhancement techniques. For instance, many previous studies have assessed object detection architectures under terrestrial benchmark datasets such as COCO or Pascal VOC but not underwater data sets which have real-world visual degradations.

Additionally, several studies have primarily been concerned with accuracy of detection, but that do not examine computational efficiency and real-time applicability. For example, in the case of underwater robotic systems often require compromise between speed and accuracy due to onboard computational resources may be limited.

Another important research gap is the lack of comparative assessment of modern transformer-based object detectors alongside traditional CNN-based architectures in underwater conditions. Although transformer-based models such as DETR and RT-DETR have recently illustrated strong performance in computer vision, their applicability for the small-object detection task in underwater environments has not been adequately explored.

This thesis aims to fill these gaps by performing extensive comparative analysis of eight state-of-the-art deep learning architectures for object detection in the underwater domain under identical experimental setups using both accuracy-based and efficiency-based evaluation metrics.

1.4 Research Objectives

The main objective of this study is to contribute underwater computer vision research by providing a comprehensive comparative analysis of advanced deep learning-based object detection models in underwater environment. The findings of this research may help researchers and developers select most suitable object detection models for underwater applications such as marine monitoring systems, autonomous underwater vehicles, intelligent robotic platforms, and underwater inspection systems.

The main objectives of this thesis are:

- To implement and analysis multiple modern deep learning-based object detection models for underwater object detection tasks.
- To evaluate detection performance using variety metrics such as mAP, precision, recall, and accuracy.
- To compare computational efficiency using inference time and FPS.
- To analyze the suitability of one-stage, two-stage, lightweight, and transformer-based architectures for underwater small-object detection by their performance.

1.5 Significance of the study

This research is significant for underwater computer vision research by providing comprehensive comparative performance analysis of modern deep learning-based object detection architectures under underwater environment. The results of this research may guide the researchers and developers to choose appropriate object detection models for underwater applications like smart robotic platforms, underwater inspection systems, marine monitoring systems and autonomous underwater vehicles. The study also contributes to the growing topics of research on object detection using transformer-based object detection and real-time underwater computer vision.

1.6 Thesis Structure

This thesis is organized into five chapters.

Chapter 1 mainly introduces the research background, problem statement, research gaps, objectives, scope, and significance of the study.

Chapter 2 highlighted the backgrounds related to underwater imaging environments, deep learning-based object detection, one-stage and two-stage detectors, transformer-based architectures, and underwater small-object detection.

Chapter 3 presents the research methodology, dataset preparation, preprocessing techniques, object detection architectures, training procedures, and evaluation metrics used in the experiments.

Chapter 4 presents the experimental results and comparative performance analysis of all object detection models.

Chapter 5 concludes the thesis and provides limitations of this study and recommendations for future research directions.

Chapter 2 Background

The last few years have seen major progress in automated object detection using artificial intelligence and computer vision. These deep learning object detection systems are employed in various applications including autonomous vehicles, robotics, medical imaging, industrial automation, surveillance, and environmental monitoring. Underwater object detection has been an emerging research area given that underwater autonomous vehicles (AUVs), remotely operated vehicles (ROVs), and intelligent marine monitoring systems (IMMS) are becoming more and more popular in underwater applications (Jian et al., 2024).

The aquatic ecosystems provide important scientific, ecological, economic and industrial information. The ability to detect objects accurately in the depths is crucial for various applications like monitoring of marine biodiversity, inspection of underwater infrastructure, ocean resources exploration, underwater archaeology and assessment of marine pollution. In contrast to terrestrial object detection, the underwater imaging environment is much more challenging for object detection as there are severe visual distortion in the undersea environment that have negative impacts on image quality and object visibility (Jaffe, 2002; Jian et al., 2024).

Deep learning has become a subfield of machine learning that has recently emerged as very successful in the ability to identify complex visual representations directly from images (LeCun et al., 2015). The performance of modern object detection models like YOLO, Faster R-CNN, RetinaNet, DETR, and RT-DETR has been seen as very successful in detecting objects under challenging visual conditions (Redmon et al., 2016; Ren et al., 2015; Carion et al., 2020; Zhao et al., 2024). The effectiveness of their applications in underwater environments still needs to be extensively studied, especially for small object detection.

This chapter presents the theoretical and technical background study of underwater object detection which is relevant to this thesis. This research study have discussed variety areas such as underwater imaging environment, and the deep learning technology in computer vision, as well as explaining the object detection method, one-stage and two-stage detectors, transformer-based detection method, small object detection problem, evaluation indicators, and comparative analysis of underwater object detection.

2.1 Underwater Imaging Environment

The imaging environment under water is very different from the imaging environment in the air due to the complexity of how light is propagated in water. Additionally, water absorbs and scatters light quickly and image visibility is decreased, as well as object boundaries are blurred, contrast is reduced, and colors are distorted. These challenges have a large impact on the functionality of computer vision algorithms and object detection systems. The author has stated that the absorption and scattering phenomena under water are major factors affecting underwater optical imaging (Jaffe, 2002). The amount of visual information that can be seen decreases sharply with increasing imaging distance. Furthermore, water have absorbed different kinds of wavelengths at different

rates that cause underwater images to appear predominantly bluish or greenish because red wavelengths disappear quickly in the underwater. Also scattering effects caused by suspended particles further degrade image quality, as a result it introduces haze-like noise and reducing object sharpness. Another author Bryson et al. (2016) assessed that underwater images often contain low illumination, dynamic backgrounds, non-uniform lighting, and motion blur which also making reliable feature extraction very complicated. In underwater environments, visual degradation becomes even more problematic when comes to detecting small objects accurately. Small underwater objects images have very few pixels and it may become partially invisible due to turbidity and low visibility conditions. Kisantal et al. (2019) pointed out that the performance of small object detection drops significantly if the features of the object are made weak in the visual system or if the spatial size of the object is reduced. In addition, underwater scenes can be visually confusing due to the presence of rocks, plants, reflections, floating particles, and marine textures that could be confused with the target objects. Such properties cause high number of false detections and lower localization accuracy. Given these difficulties, it is very important to make sure that the models have a strong feature-learning mechanism that can learn from degraded visual environments and extract meaningful information from them for underwater object detection. Moreover, advanced deep learning-based detection architectures have demonstrated their potential capabilities in solving such problems through hierarchical feature learning and contextual representation (LeCun et al., 2015).

2.2 Computer Vision based detection applications

Deep learning model has become one of the most prominent technologies in modern computer vision research areas. Previously, traditional image processing methods relied on handcrafted feature extraction methods, but modern deep learning models can automatically learn hierarchical visual representations directly from massive datasets in a short time. Most computer vision systems today that use deep learning rely on Convolutional Neural Networks (CNNs). CNN architectures use convolutional layers for learning spatial patterns, textures, edges, and semantic visual features from images. The authors have described that deep learning enables end-to-end representation learning which can greatly boost performance in image classification, object detection and segmentation tasks (LeCun et al., 2015). After the region-based CNN architectures and real-time object detectors were introduced, deep learning has shown its effectiveness in object detection. In the modern era, object localization and object classification can be done simultaneously with high accuracy by deep learning based detectors. Underwater environments have very challenging visual conditions due to the wide variation in the light scattering in the water, making it a highly suitable use for deep learning in underwater object detection. CNN-based feature learning can be used to automatically obtain strong visual features even in poor imaging conditions. In recent years, transformer-based deep learning architectures are also becoming great alternatives to the traditional CNN-based systems. Transformers rely on attention mechanisms to take into account spatial relations between different parts of an image over long-range distances and also it is enhancing the ability to understand and represent the image contextually (Carion et al., 2020; Zhao

et al., 2024). Dosovitskiy et al. (2021) showed that transformer-based vision architectures were effective for image recognition and that they later had an impact on object detection systems like DETR and RT-DETR.

Object detection is a basic problem in computer vision task that involves identifying types of object categories and localizing their positions from the images. object detection is complex task than image classification, because it requires both recognition and precise spatial localization of multiple objects simultaneously. In general, modern object detection systems normally generate bounding boxes around detected objects and then classify the object and assign confidence scores. Another author have assessed that object detection combines feature extraction, classification, and localization into a unified visual recognition framework (Szeliski, 2022). Furthermore, deep learning-based object detection methods are generally divided into two major groups where one group is called one-stage detectors, and another one is two-stage detectors. One-stage detectors are detectors which localize and classify objects in one single forward pass of the network. These models are typically designed to run in real time at high speeds. For example, One-stage detectors including YOLO, SSD, and RetinaNet architectures. Two-stage detectors have two stages where it first generates candidate object proposals and then classify and refine those proposals. These architectures generally provide better localization accuracy, but it needs more computational cost. These are some of the two-stage detectors such as Faster R-CNN, Mask R-CNN. Very recently, another detector model called transformer-based detectors introduced as a third major paradigm in object detection research. Transformer detectors introduce several handcrafted components in the detection pipeline and eliminate the need for global contextual relationships in images from the detection model by employing self-attention. In underwater systems, automatic scene understanding is essential for underwater robotic platforms for navigation, monitoring, and inspection. The automatic object detection is a particularly important role in underwater systems.

2.3 Detection Models

One-stage object detectors simultaneously localize and classify objects in a single architecture of a neural network. These models directly learn object classes and coordinates of bounding boxes from feature maps without an extra region proposal phase. One-stage detectors have the most superior efficiency in making inferences and can be used in real time. The author Redmon et al. (2016) stated that one-stage detection frameworks significantly improve computational efficiency by simplifying the detection pipeline. The YOLO (You Only Look Once) family represents one of the most influential one-stage detection frameworks. YOLO architectures divide images into grids and predict bounding boxes directly in a single forward pass. Modern YOLO versions such as YOLOv8 incorporate anchor-free prediction mechanisms and improved feature extraction techniques (Jocher et al., 2023). SSD (Single Shot MultiBox Detector) is another important one-stage detection architecture designed for efficient multi-scale object detection. SSDlite extends SSD by integrating lightweight MobileNet backbones suitable for mobile and embedded devices.

RetinaNet introduced Focal Loss to address foreground-background imbalance problems commonly observed in dense object detection. According to Lin et al. (2017), Focal Loss improves training stability by reducing the influence of easy negative samples during optimization. One-stage detectors are highly relevant for underwater robotic systems because many underwater applications require real-time image processing with limited computational resources.

Two-stage object detectors are object detectors that break the object detection process into two stages. The first stage identifies the candidate object regions, and the second stage categorizes and narrows down the candidate regions. Faster R-CNN is among the most crucial two-stage object detection architectures. The model incorporated the Region Proposal Network (RPN), and made the proposal generation process far more efficient than previous region-based models. Ren et al. (2015) proposed an R-CNN system called Faster R-CNN that directly incorporated the proposal generation steps into the convolutional network architecture to achieve the best object detection accuracy. Mask R-CNN extended Faster R-CNN to include an additional segmentation branch to segment the object at the instance level. While for segmentation, Mask R-CNN showed to be very accurate at object detection as well, because of its general ability to represent the features. Two-stage detectors are of special interest in the underwater small object detection, where the two proposals refinement stages allow the precise localization of the visually blurred or occluded object. These architectures, however, typically require greater resources to run the computations and slower time to make an inference than the lightweight one-stage detectors.

Another detector model, the architectures of transformers were initially developed for natural language processing, but were heavily influential to computer vision research. Transformer-based object detectors employ self-attention mechanisms to learn the overall context relationship between image regions. For example, DETR (Detection Transformer) proposed a novel end-to-end object detection framework without anchor boxes and non-maximum suppression operations of traditional detectors. According to Carion et al. (2020), DETR reformulated object detection as a direct set of prediction problem by use of transformer encoder-decoder architectures. Later, RT-DETR expanded the transformer-based detection framework to real-time applications via introducing efficient feature aggregation mechanisms and computing architectures (Zhao et al., 2024). Underwater applications are of special interest due to the fact that attention mechanisms may become more robust to detect a target in cluttered background and visually degraded situations when the detector is based on a transformer. But the need for more computational resources and longer training time is a common issue with transformer architectures compared to lightweight CNN-based detectors.

2.4 Evaluation Metrics in Object Detection

It is difficult to evaluate the performance of object detection systems using a single criterion, as they must meet the criteria of accuracy in localization, reliability of the classification and efficiency of computation. One of the most common evaluation methods used in the field of object detection is called Mean Average Precision (mAP). Everingham et al., (2010) described that the

mAP as a measure that represents the relationship between precision and recall, and that measures localization accuracy and classification reliability. This thesis uses both $\text{mAP}@0.5$ and $\text{mAP}@0.5:0.95$, where $\text{mAP}@0.5$ assesses the localization accuracy of basic accuracy and $\text{mAP}@0.5:0.95$ is the strict localization assessment with multiple IoU thresholds. Other metrics like precision, recall, detection accuracy, frames per second (FPS), inference time, and parameter count were also selected as they are important for underwater systems which need to balance the accuracy and the computational efficiency. Underwater applications are sensitive to real-time data, which demand a high inference speed along with high accuracy in object localization. In addition, multiple evaluation metrics such as precision, recall, and Intersection over Union (IoU) are required for comprehensive assessment of detection performance in object detection in the underwater environment (Padilla et al., 2020).

2.5 Previous Studies and Challenges in Underwater Object Detection

Recently, there have been several studies that examined deep learning based underwater object detection systems. But most of the works are targeted on specific architectures and not a detailed comparative analysis of architectures for multiple detection paradigms. Underwater image degradation often results from various factors, and the performance of feature extraction is crucial for dealing with degradation and low-visibility scenarios (Jian et al., 2024). Transformer-based models have been recently investigated for other underwater visual tasks, and still large-scale comparisons are limited. Most existing research either emphasizes either accuracy of detection or even only efficiency of computation. However, comprehensive comparison across different detection paradigms is still limited in underwater environment as indicated by recent surveys (Chen et al., 2024). Small object detection is one of the challenging problems in object detection research. In addition, the spatial information of small objects is limited, and important visual features are lost during the convolutional down sampling operation. Kisantal et al. (2019) argue that small object detection is plagued with low quality of feature representation, problem of localization and class imbalance. It is even more challenging in the underwater environment, due to degradation in the underwater images and the poor quality of the visual features. Small objects under water can seem blurred, obscured, or merged with the background. Modeling small objects is a challenging task since model architectures have been widely investigated to enhance the performance in small-object detection. A wide range of model architectures have been studied to boost the performance in small-object detection by Feature Pyramid Networks (FPNs), attention mechanisms, multi-scale feature learning, and Transformer-based contextual modeling. The object categories used in this thesis, such as propellers, pipes, QR codes, and nets, which are frequently captured as small or partially visible objects in underwater images. Therefore, small-object detection capability is highly important for this research.

This chapter mainly presented the theoretical and technical background which are related to underwater object detection and deep learning-based computer vision systems. The discussion highlighted different areas such as underwater imaging environments, deep learning in object detection, one-stage and two-stage detection architectures, transformer-based detectors, small-

object detection challenges, and evaluation methodologies. The reviewed literature demonstrates that underwater object detection still remains a highly challenging research task because of its severe visual degradation and lack of computational resources. Existing studies also reveal that very limited number of comparative evaluations have been completed between modern object detection paradigms under identical underwater experimental conditions.

The next chapter presents research methodology, dataset preparation process, preprocessing techniques, object detection architectures, implementation details, and evaluation strategies used in this thesis.

Chapter 3 Research Methodology and Experimental Design

3.1 Introduction

This chapter presents the complete research methodology used in this thesis for underwater small-object detection using deep learning-based object detection models. The chapter explains the overall experimental design, dataset preparation process, preprocessing strategy, implementation environment, model selection criteria, model architectures, training procedures, and evaluation metrics used throughout the study. The purpose of combining these methodological components into a single chapter is to provide a complete understanding of how the experiments were designed, implemented, and evaluated in a single conceptual framework.

Most of the challenges in underwater object detection are recognized as visual degradations including low visibility, light absorption, color distortion, suspended particles, motion blur, complex background and so on. Light attenuation and scattering in water cause these degradations and degrade the quality of images and objects in water (Jaffe 2002; Jian et al. 2024). Especially small objects in the water can be hard to find since they only take up a small number of pixels and can be partially occluded or have a very similar appearance to the background. This is well-known in the field of small-object detection studies, where the few visual features lead to smaller detection accuracy (Kisantal et al., 2019). Therefore, the proper choice of deep learning networks and assessment methods is crucial to guarantee accurate underwater detection results.

This research evaluates eight modern deep learning-based object detection models, including YOLOv8n, YOLOv10n, RT-DETR-L, Faster R-CNN, RetinaNet, SSDlite320-MobileNetV3, Mask R-CNN ResNet50-FPN, and DETR ResNet-50. These models have highlighted different paradigms of object detection systems, including one-stage detectors, two-stage detectors, and transformer-based architectures (Redmon et al., 2016; Ren et al., 2015; Carion et al., 2020; Zhao et al., 2024). Evaluating these models under the same identical experimental conditions allows this study a meaningful comparison between detection accuracy, computational efficiency, inference speed, and real-time applicability in underwater environments.

3.2 Research Methodology

This study have used quantitative experimental research method. The study compared several deep learning-based object detection models on the same data set under the same training set-up, data pre-processing technique, and evaluation protocol. All comparative experiments were done under the same experimental conditions, thus ensuring fairness and consistency.

The research process first started with underwater dataset preparation and annotation verification. After that, the dataset was then divided into two subsets such as training set and testing set. Each object detection model was trained separately with the same partitions of the data and using the same evaluation method. After the training, the models were tested according to the common

object detection performance evaluation metrics including $mAP@0.5$, $mAP@0.5:0.95$, precision, recall, detection accuracy, FPS, training time and inference time.

Lastly, in this research work, the comparative analysis was carried out to analyze the pros and cons of each model in the small-object detection task in underwater environment. The experimental procedure of this study is presented in the next chapter 4 Results and Analysis.

3.3 Dataset Preprocessing and Experimental Environment

This thesis used an underwater object detection dataset comprising six object classes that are frequently seen in underwater environments for the experiments. The object categories included propeller, pipe_type2, red_fin, net, qr_codes, and pipe. The dataset have underwater images which were captured under specific environmental conditions such as low illumination, underwater turbidity, color attenuation, object occlusion, and cluttered backgrounds.

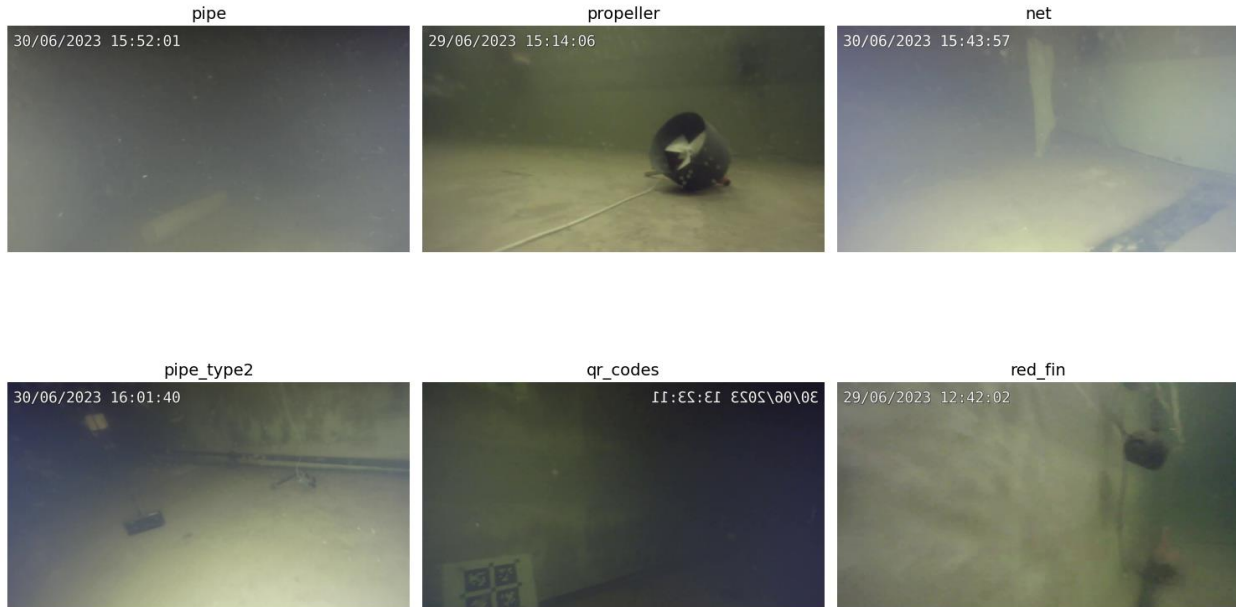


Figure 3. 1: Underwater Detection Dataset Object Categories

The dataset was split into a training and testing set to evaluate the model fairly. The training data set was used to learn the model and optimize the parameters, and the testing data set was used only for evaluation of the detection performance after the model is trained.

In the dataset, all of the object annotations were stored in YOLO text-based annotation format. Every annotation file has normalized bounding-box coordinates along with class identifiers corresponding to the underwater object categories. During the implementation, these normalized coordinates were later transformed into absolute pixel coordinates for compatibility with PyTorch-based detection models.

The underwater dataset applied in this study is especially appropriate for the assessment of small-object detection performance since many of the objects to be detected are relatively small image areas. Small-object detection is still a challenge because of the limited representation of features and the reduction of visibility of the objects (Kisantal et al., 2019).

Furthermore, the next step is data preprocessing which plays a significant key role in deep learning-based object detection because improper image formatting or annotation inconsistency can negatively impact on the feature extraction and model convergence. In this study, all underwater captured images were converted into RGB format before training. Then, the images were converted into tensor representations during stage of the data-loading. During the data-loading stage, YOLO annotation labels were converted into bounding-box coordinate formats compatible with PyTorch and Ultralytics detection frameworks.

The PyTorch DataLoader framework was employed to ensure efficient computation and memory utilization for the training process, as data was loaded in batches. Data was loaded in batches for efficient computation and memory utilization during training using the PyTorch DataLoader framework. The training dataset was shuffled to enhance the generalization ability and to prevent overfitting. To fairly compare the models, the preprocessing strategy was carefully kept the same. Aggressive image enhancement techniques or any synthetic augmentation was not applied as this study wanted to compare the inherent capabilities of the selected object detection architectures under the same underwater conditions in the objective of this study. Experiments were conducted with the Python-based deep learning frameworks PyTorch, Torchvision and Ultralytics. Cloud computing platforms like Google Colab and Kaggle Notebook were used to enable GPU acceleration via CUDA-supported environments. During implementation, to process the images, address numerical operations, and visualize the results, several supporting libraries were employed, such as OpenCV, NumPy, Matplotlib, and TorchMetrics. The adoption of GPU acceleration was necessary as today's object detection models for deep learning algorithms have a large number of trainable parameters, and training and inference demands significant computational resources. According to Goodfellow et al. (2016), GPU-based parallel processing significantly improves the deep learning training efficiency and enables the practical implementation of large-scale neural network architectures.

3.4 Model Selection Strategy

In this thesis, the object detection models have selected based on several important considerations, especially based on detection accuracy, real-time capability, architectural diversity, computational efficiency, and popularity in recent computer vision research.

The selected models represent different categories of object detection systems where YOLOv8n and YOLOv10n represent lightweight one-stage detectors optimized for real-time performance. Other models like Faster R-CNN and Mask R-CNN represent classical two-stage detection architectures known for high localization accuracy. RetinaNet is a dense one-stage detector that is developed to solve the foreground-background imbalance problems while SSDlite320-

MobileNetV3 models are lightweight mobile-oriented detection systems, which can be deployed on edge devices. Additionally, RT-DETR-L and DETR ResNet-50 highlights transformer-based detection architectures which are capable of learning global contextual relationships within underwater images. These various architectures enable comprehensive performance comparison between modern object detection paradigms under underwater imaging environment. Another author has stated that one-stage detectors provide high-speed inference (Redmon et al. (2016), while two-stage detectors achieve stronger localization accuracy (Ren et al., 2015). Additionally, another updated detector like Transformer-based architecture can further improve global contextual understanding through attention mechanisms (Carion et al., 2020; Zhao et al., 2024). Therefore, evaluating these model families together gives valuable insight into the trade-offs between the accuracy and the computational efficiency of these models.

3.4.1 YOLOv8n Architecture

YOLOv8n is a lightweight one-stage object detection model which is developed by Ultralytics. The model belongs to the YOLO (You Only Look Once) family of real-time object detectors that are intended to perform high speed object localization with relatively low computational complexity. It is different to previous YOLO versions, where YOLOv8 introduces anchor-free detection and improved feature-learning mechanisms for real-time object detection (Jocher et al., 2023). This architecture uses a CSP-based backbone together with feature pyramid structures for multi-scale feature learning. The model also incorporates optimized prediction heads that improve localization performance while preserving inference speed. YOLOv8n was selected in this study because underwater robotic systems and marine monitoring applications often require real-time object detection capabilities. The lightweight design of YOLOv8n enables fast inference while maintaining strong detection accuracy for small underwater objects in deep-sea settings. The authors expressed that YOLOv8 architectures improved speed-accuracy trade-offs compared to earlier YOLO generations and it is highly suitable for edge-device deployment and real-time applications (Jocher et al., 2023).

3.4.2 YOLOv10n Architecture

YOLOv10n is one of the latest developments in the YOLO family and introduces several architectural improvements designed to reduce computational redundancy and improve inference efficiency. The model introduces consistent dual assignments and end-to-end optimization strategies that reduce post-processing operations and improve real-time performance. Compared to previous YOLO architectures, YOLOv10 minimizes unnecessary computations during inference while maintaining strong detection accuracy. YOLOv10n was included in this thesis because of its potential suitability for underwater real-time systems where low latency and efficient processing are important operational requirements. According to Wang et al. (2024), YOLOv10 achieves superior efficiency through optimized detection pipelines and balanced architectural scaling.

3.4.3 RT-DETR-L Architecture

RT-DETR-L is a transformer-based real-time object detector developed to combine the global contextual reasoning capability of transformers with practical inference speed. Traditional transformer-based detectors often require large computational resources and long inference time. RT-DETR addresses these limitations through efficient hybrid encoder designs and optimized decoder structures. The architecture uses transformer attention mechanisms to learn long-range spatial relationships between image regions. This capability may be particularly useful in underwater environments where objects appear partially occluded or visually degraded. RT-DETR-L was selected because transformer-based detectors have recently shown strong performance in complex object detection tasks involving cluttered backgrounds and small objects. According to Zhao et al. (2024), RT-DETR achieves state-of-the-art real-time transformer-based object detection performance while preserving high localization accuracy.

3.4.4 Faster R-CNN Architecture

Faster R-CNN is a classical two-stage object detection architecture proposed by Ren et al. (2015). The model first generates candidate object proposals using a Region Proposal Network (RPN), after which the proposals are classified and refined. The architecture is widely recognized for its strong localization accuracy and robust feature representation capability. Faster R-CNN performs particularly well in scenarios involving small objects and complex backgrounds because the two-stage pipeline enables more precise bounding-box refinement. Faster R-CNN was included in this study because it remains one of the most important benchmark models in object detection research and provides strong comparison against newer lightweight detectors. Authors have demonstrated that the integration of region proposal generation directly into the convolutional network significantly improves detection efficiency compared to earlier region-based approaches (Ren et al. (2015).

3.4.5 RetinaNet Architecture

RetinaNet is another one-stage object detector which is designed for addressing the foreground-background imbalance problem that is commonly observed in dense object detection tasks. This architecture developed Focal Loss, which reduces the influence of easy negative samples during training as well as force the model to focus more on complex objects. Furthermore, RetinaNet combines with a ResNet backbone with Feature Pyramid Networks (FPN) which is improving multi-scale feature extraction. This multi-scale learning capability is significantly important for underwater small-object detection because underwater objects frequently appear at different scales and resolutions in images. According to the authors, Lin et al. (2017) have showed that Focal Loss significantly improves dense object detection accuracy through addressing their class imbalance problems during optimization.

3.4.6 SSDlite320-MobileNetV3 Architecture

In this study, SSDlite320-MobileNetV3 model is used because it is a lightweight object detection model specifically designed for mobile and embedded systems with limited computational resources. This architecture combines with the SSD (Single Shot Multi-Box Detector) framework with the efficient MobileNetV3 backbone because MobileNetV3 uses depth wise separable convolutions and lightweight optimization techniques to reduce computational complexity. Another reason, the SSDlite320-MobileNetV3 architecture was selected because underwater robotic systems often operate on hardware platforms with limited processing power and energy constraints. Therefore, lightweight detection models are important for practical deployment scenarios. In this related topic, the authors have discussed that MobileNetV3 achieves efficient mobile vision performance through neural architecture search and lightweight network optimization (Howard et al., 2019).

3.4.7 Mask R-CNN ResNet50-FPN Architecture

Mask R-CNN is the important architecture in this research which extends Faster R-CNN architecture by introducing an additional branch for instance segmentation together with small object detection. This architecture can perform object classification, bounding-box regression as well as segmentation mask prediction simultaneously. Although this thesis focuses only on object detection performance, Mask R-CNN was applied because it represents an advanced two-stage detection framework with strong feature representation capability for large datasets. Also, this ResNet50-FPN model improves the multi-scale feature extraction and enables more accurate results for small underwater objects. Previous study has highlighted that Mask R-CNN preserves high object detection accuracy while enabling flexible pixel-level segmentation capability (He et al., 2017).

3.4.8 DETR ResNet-50 Architecture

DETR (Detection Transformer) is a transformer-based object detection architecture which is also used in this study that formulates object detection as a direct set prediction problem. This model is different than other traditional object detectors because DETR model removes the need for anchor boxes and non-maximum suppression. The main improvement in this architecture is that it combines CNN-based feature extraction with transformer encoder-decoder attention mechanisms. The global attention capability of DETR enables the model to learn contextual relationships between image regions, which may improve understanding of cluttered underwater scenes. Here in this study, DETR ResNet-50 was included because transformer-based object detectors represent an important research direction in modern computer vision. According to Carion et al. (2020), DETR simplifies the object detection pipeline while this model gaining competitive detection performance through transformer attention mechanisms.

3.4.9 Comparative Expectations Before Experimentation

Before implementing the experiments, several performance expectations were established based on the architectural characteristics of the eight selected object detection models which were used in this study, namely YOLOv8n, YOLOv10n, RT-DETR-L, Faster R-CNN, RetinaNet, SSDlite320-MobileNetV3, Mask R-CNN ResNet50-FPN, and DETR ResNet-50. YOLOv8n and YOLOv10n were expected to achieve strong real-time performance together with high detection accuracy because they are lightweight one-stage detectors optimized for efficient inference while RT-DETR-L and DETR ResNet-50 were expected to provide strong detection capability and contextual understanding due to their transformer-based architectures. However, these models were also expected to require higher computational resources and longer inference time due to its capabilities. On the other hand, Faster R-CNN and Mask R-CNN ResNet50-FPN were expected to achieve high localization accuracy because of their two-stage detection mechanisms and strong feature extraction capability, although their processing speed was anticipated to be slower than lightweight one-stage detectors. Another important model that is RetinaNet which was expected to provide balanced performance between speed and accuracy because of its Feature Pyramid Network (FPN) structure and Focal Loss optimization strategy which has designed for dense object detection. Lastly, SSDlite320-MobileNetV3 architecture was expected to achieve the fastest inference speed and lowest computational cost due to its lightweight mobile-oriented architecture, although its detection accuracy was expected to be comparatively lower than larger and more complex models. So, these expectations were systematically evaluated and verified through the experimental analysis and their comparative performance assessment which were presented in the next chapter.

3.5 Evaluation Metrics

It is important to evaluate the object detection models, because a single metric cannot thoroughly represent the performance of a detection system. Object detection in underwater environments is a complex process that is plagued with several difficult problems, including poor localization, omission, false detection and computational constraints. Therefore, multiple evaluation metrics are used to provide a comprehensive and balanced assessment of detection systems across accuracy and efficiency dimensions (Everingham et al., 2010; Padilla et al., 2020). In this study, the selected metrics include Mean Average Precision ($mAP@0.5$ and $mAP@0.5:0.95$), Precision, Recall, Detection Accuracy, Frames Per Second (FPS), Average Inference Time, Total Inference Time, and Parameter Count to evaluate the performance of a detection system.

The use of different kinds of evaluation metrics allows comprehensive comparison between the selected deep learning models under the same underwater conditions. These metrics are widely used in previous object detection study because they provide balanced evaluation of localization quality, classification reliability, speed, and deployment feasibility. The authors Everingham et al., (2010) argued that object detection evaluation must simultaneously measure both localization

correctness and classification performance because successful detection requires accurate object classification as well as correct bounding-box position.

3.5.1 Intersection over Union (IoU)

Intersection over Union (IoU) is computed first then the main detection metrics are calculated. The IoU indicates how many of the pixels of the predicted bounding box are covered by the ground-truth

The IoU value is calculated as:

$$IoU = \frac{\text{Area of Overlap}}{\text{Area of Union}} \text{-----}(1)$$

where:

- Area of Overlap indicates the common area between the predicted and ground-truth bounding boxes.
- Area of Union represents the total area which is covered by both bounding boxes.

The IoU value ranges starts from 0 to 1. An IoU value closer to 1 indicates the better localization accuracy of object detection. In object detection study, a prediction is generally counted as correct if the IoU exceeds a predefined threshold such as 0.5. IoU is particularly important in underwater object detection because underwater objects often appear partially occluded, blurred, or distorted due to complex water conditions. Therefore, accurate localization evaluation is necessary to determine whether the predicted bounding boxes properly capture underwater objects. One of the previous studies, the authors have highlighted that IoU is one of the most fundamental localization metrics used in modern object detection (Rezatofghi et al., 2019).

3.5.2 Mean Average Precision (mAP)

Mean Average Precision (mAP) is one of the most important and widely used evaluation metrics in object detection research study because it measures both localization performance and classification accuracy simultaneously at the same time.

In this object detection tasks, first precision and recall values are computed at different confidence thresholds. After that, the Precision-Recall curve is generated, and then the Average Precision (AP) is calculated as the area under the Precision-Recall curve for each object class.

In object detection, Average Precision (AP) is calculated as:

$$AP = \int_0^1 \text{Precision}(\text{Recall}) d\text{Recall} \text{-----}(2)$$

After that, the most important metric Mean Average Precision (mAP) is calculated as the average of AP values across all object classes:

$$mAP = \frac{1}{N} \sum_{i=1}^N AP_i \text{-----}(3)$$

where:

- N represents the total number of object classes.
- AP_i represents the Average Precision of class i .

In this research, two mAP metrics were used. First using mAP@0.5 metric which evaluates the detection performance by using an IoU threshold of 0.5. A predicted object is considered as correct when the overlap between the predicted bounding box and the ground-truth box is at least 50%. This metric is commonly used because it provides straightforward interpretation of detection capability and localization correctness.

Then the study used another metric mAP@0.5:0.95 which is a stricter evaluation metric that averages AP values across multiple IoU thresholds ranging from 0.5 to 0.95 with increments of 0.05.

The metric can be represented as:

$$mAP_{0.5:0.95} = \frac{1}{10} \sum_{IoU=0.5}^{0.95} AP_{IoU} \text{-----}(4)$$

This metric provides more comprehensive evaluation of localization precision because it penalizes inaccurate bounding-box predictions more strongly.

The reason for using both mAP@0.5 and mAP@0.5:0.95 in this thesis is that underwater object detection requires not only successful object identification but also accurate localization under visually degraded conditions. The mAP@0.5:0.95 metric provides a stricter and more comprehensive evaluation because it evaluates performance across multiple localization thresholds (Padilla et al., 2020).

3.5.3 Precision

Precision basically measures the reliability of positive detections generated by the model. It evaluates how many predicted objects are actually correct.

Precision is calculated using Equation:

$$Precision = \frac{TP}{TP+FP} \text{-----}(5)$$

where:

- TP represents as True Positives.
- FP represents as False Positives.

Precision reflects how reliably the model can predict objects in the model and it is especially important in environments where false detections can be misleading to the overall architectures (Sokolova and Lapalme, 2009). This metric is important in underwater complex environments because underwater conditions often contain visually confusing structures such as floating particles, reflections, rocks, and marine textures that may lead to false object predictions. A model with high precision is considered more reliable because most of its detections are correct. The author Sokolova and Lapalme (2009) described that precision is especially important when false detections can negatively affect system reliability and operational safety.

3.5.4 Recall

Recall measures the ability of the model to detect all existing objects within an image.

Recall is calculated using Equation as:

$$Recall = \frac{TP}{TP+FN} \text{-----}(6)$$

False Negatives represent missed detections in the architecture. High recall indicates that the model successfully identifies most underwater objects without failing to identify the objects correctly. This metric is important because missing underwater objects may negatively affect in applications such as underwater robotic navigation, marine monitoring systems, and inspection operations. The recall performance for the detection of small objects in underwater images is especially sensitive due to the difficulty of seeing and recognizing small objects under low-visibility conditions and fading image quality. According to Padilla et al., (2020), recall is critical for evaluating the completeness of object detection systems.

3.5.5 Detection Accuracy

Detection Accuracy measures the overall proportion of correct detections relative to the total number of predictions.

Detection Accuracy is calculated using Equation as:

$$Detection\ Accuracy = \frac{Correct\ Detections}{Total\ Predictions} \times 10 \text{-----}(6)$$

This is a measure of overall interpretation of the correctness of the object detection system. While mAP is more complete in its evaluation, detection accuracy summarizes the overall effectiveness of each of the models in correctly detecting underwater objects. This metric is particularly useful

because it allows for intuitive understanding of the reliability of the model built, and can be used for comparison across different model architectures.

3.5.6 Frames Per Second (FPS)

Frames Per Second (FPS) measures the real-time processing capability of the object detection model.

FPS is calculated using Equation as:

$$FPS = \frac{Total\ Number\ of\ Test\ Images}{Total\ Inference\ Time} \text{-----}(7)$$

FPS is critical for real-time applications and reflects the deployment feasibility of detection systems (Bochkovskiy et al., 2020). Higher FPS values indicate the faster object detection performance of a detection model. FPS is an important metric in underwater applications because real-time underwater detection systems require fast image processing capability. Real-time underwater detection systems must process incoming video streams quickly while maintaining acceptable detection accuracy. According to Bochkovskiy et al. (2020), real-time object detection systems require balanced optimization between speed and accuracy.

3.5.7 Average Inference Time

Average Inference Time measures the average time required by the model to process a single batch or image during testing.

The metric is calculated using Equation:

$$Average\ Inference\ Time = \frac{Total\ Inference\ Time}{Number\ of\ Test\ Batches} \text{-----}(9)$$

Lower inference time indicates the capability of prediction faster of a model. Inference time is important because underwater real-time systems often operate under computational limitations and energy constraints. In this case, faster inference can improve responsiveness and enables practical deployment in embedded underwater systems.

3.5.8 Total Test Inference Time

Total Test Inference Time measures the total amount of time required to process the overall testing dataset.

The metric is calculated using Equation:

$$Total\ Test\ Inference\ Time = \sum_{i=1}^N Inference\ Time_i \text{ -----}(10)$$

where:

- N represents the total number of test batches or images.

This measure is a reflection of computational expense for large-scale deployment and gives overall idea of computational workload and practicability of deployment. In general, the models with lower total inference time are better for large-scale underwater monitoring applications.

3.5.9 Parameter Count

Parameter Count represents the total number of trainable parameters within a deep learning model.

The metric is calculated using Equation:

$$Parameter\ Count = \sum_{i=1}^N Trainable\ Parameter\ s_i \text{ -----}(11)$$

Models with larger parameter counts generally have stronger feature-learning capability but require very large amount of computational cost and memory usage. Models with fewer parameters are generally more suitable for deployment on limited resource devices (Howard et al., 2019). However, lightweight models may sometimes sacrifice localization accuracy for computational efficiency. Thus, it is crucial to analyze the number of parameters to understand the relationship between model complexity and detection performance. According to Howard et al. (2019), lightweight architectures are essential for efficient deployment on mobile and embedded systems.

3.5.10 Importance of Multi-Metric Evaluation

Multiple evaluation metrics are used to ensure a comprehensive evaluation of the accuracy, localization precision, and computation efficiency of object detection (Everingham et al., 2010; Padilla et al., 2020). For instance, mAP metrics assess the quality of localization and classification, precision indicates the reliability of prediction, recall indicates the object coverage capability, detection accuracy reflects the overall correctness, and FPS together with inference time reflects computational efficiency. Using only a single metric could produce misleading conclusions because some models may achieve high accuracy but poor real-time performance, while others may operate very quickly but sacrifice localization quality, so it is very important to use multiple evaluation metrics. Furthermore, combining with accuracy-based and efficiency-based metrics provides comprehensive understanding of the practical strengths and limitations of each underwater object detection architecture.

Chapter 4 Results and Analysis

4.1 Chapter Overview

This chapter describes the experimental results of the underwater object detection models which are evaluated in this thesis study. In this research, all the models were compared using the same performance indicators which have already explained the following previous chapter methodology and experimental setup. These include $mAP@0.5$, $mAP@0.5:0.95$, precision, recall, detection accuracy, parameter count, training time, average inference time, total test inference time, and frames per second. The purpose of this chapter is not only to focus on which model achieved the strongest performance, but also to consider both detection quality and computational efficiency in terms of identifying the most suitable model for underwater small-object detection.

The final comparative analysis includes eight models such as YOLOv8n, YOLOv10n, RT-DETR-L, Faster R-CNN, RetinaNet, SSDlite320-MobileNetV3, Mask R-CNN ResNet50-FPN, and DETR ResNet-50. These models represent lightweight one-stage detectors, heavier two-stage detectors, transformer-based detectors, and mobile-oriented models. This wide selection of model makes it possible to compare different design philosophies under the same underwater dataset and shared same evaluation matrices.

This chapter is organized into four main parts. First, the overall quantitative comparison is presented in two tables. Second, the results are analyzed using the main detection quality metrics. Thirdly, computational efficiency and deployment suitability are discussed. Finally, this chapter explains the results based on research questions and shows which models are most suitable for different underwater applications.

4.2 Overall Quantitative Results

To improve readability in this thesis paper, the overall comparison is divided into two tables. In the first Table 7.1 focuses on detection quality, while the second Table 7.2 presents computational and efficiency-related results.

Table 4. 1 Overall Comparison of Detection Performance

Model	$mAP@0.5$ (%)	$mAP@0.5:0.95$ (%)	Precision (%)	Recall (%)	Detection Accuracy (%)
YOLOv8n	98.46	81.85	96.06	95.40	96.06
YOLOv10n	96.84	80.99	94.63	89.20	94.63

Model	mAP@0.5 (%)	mAP@0.5:0.95 (%)	Precision (%)	Recall (%)	Detection Accuracy (%)
RT-DETR-L	97.90	82.06	94.62	97.01	94.62
Faster R-CNN	97.93	74.80	91.40	97.70	91.40
RetinaNet	98.16	79.26	92.73	96.78	92.73
SSDlite320-MobileNetV3	81.52	47.34	77.23	67.82	77.23
Mask R-CNN ResNet50-FPN	95.27	74.32	80.91	94.48	80.91
DETR ResNet-50	45.20	31.38	64.53	39.31	64.53

Table 4. 2 Overall Comparison of Computational Efficiency

Model	Parameters	Training Time (s)	Avg. Inference Time (s)	Total Test Inference Time (s)	FPS
YOLOv8n	3,157,200	931.52	0.0088	2.80	113.67
YOLOv10n	2,775,520	1212.15	0.0101	3.22	98.80
RT-DETR-L	32,970,476	5701.27	0.0416	13.22	24.05
Faster R-CNN	41,324,786	12323.82	0.3384	27.08	11.75
RetinaNet	32,272,419	10309.75	0.2818	22.54	14.11
SSDlite320-MobileNetV3	3,792,380	4362.39	0.0197	3.14	101.42
Mask R-CNN ResNet50-FPN	43,949,305	13469.76	0.1676	26.65	11.93
DETR ResNet-50	41,502,923	13175.06	0.1234	19.62	16.21

The two tables show that the models behaved very differently across the selected metrics. The lightweight YOLO models achieved both high accuracy and excellent speed. RT-DETR-L

achieved the best strict localization performance and very high recall, but it required more computation. Faster R-CNN and RetinaNet produced strong detection quality, but they were slower and heavier. Mask R-CNN produced reasonably strong detection results but was computationally expensive. SSDlite320-MobileNetV3 was highly efficient and fast, but its detection accuracy was clearly weaker than the top models. DETR ResNet-50 performed the worst both in detection quality and in practical efficiency.

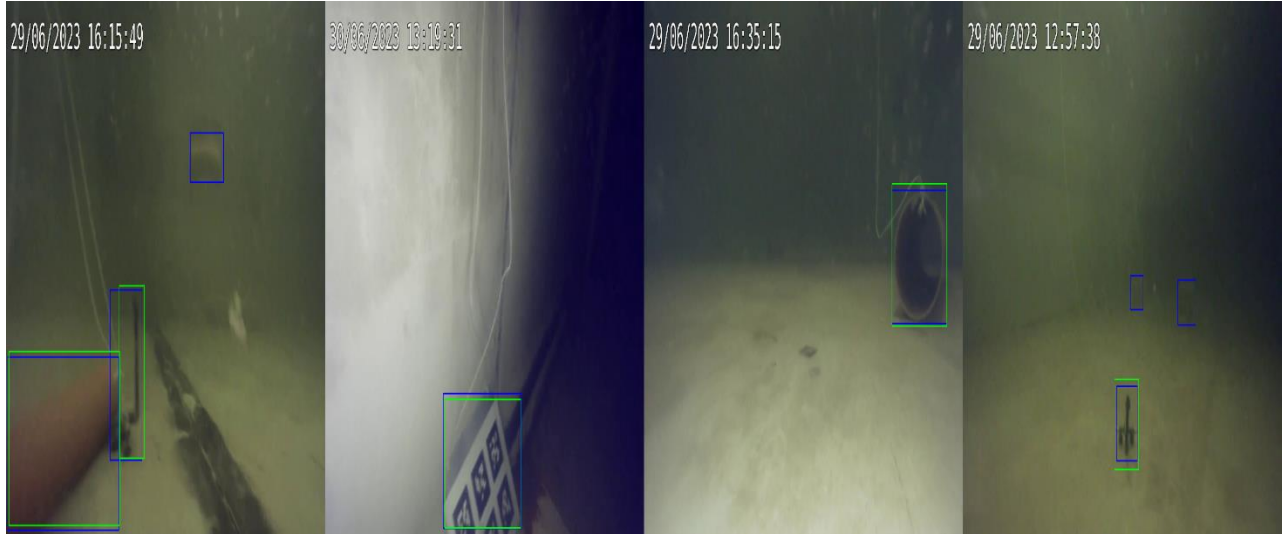


Figure 4. 1: Sample Underwater Detection Outputs

This Figure 4.1 presents representative underwater object detection outputs produced during the experimental evaluation. The images demonstrate the ability of the evaluated models to detect and localize small underwater objects under challenging visual conditions including low illumination, turbidity, cluttered backgrounds, and color distortion. The detection outputs illustrate that several objects such as pipes, QR markers, and underwater structures were successfully localized despite degraded underwater visibility. These qualitative examples support the quantitative evaluation results discussed in this chapter and provide visual interpretation of the detection performance of the models in realistic underwater environments.

4.3 Comparison Based on Detection Quality

4.3.1 mAP@0.5

The first major metric used in the comparison is mAP@0.5. This metric measures how well the models detect and localize objects using a standard IoU threshold of 0.5. It provides a widely accepted evaluation of detection performance by combining classification and localization

accuracy (Everingham et al., 2010). As described in the experimental setup chapter, this metric was one of the central performance indicators used throughout the experiments.

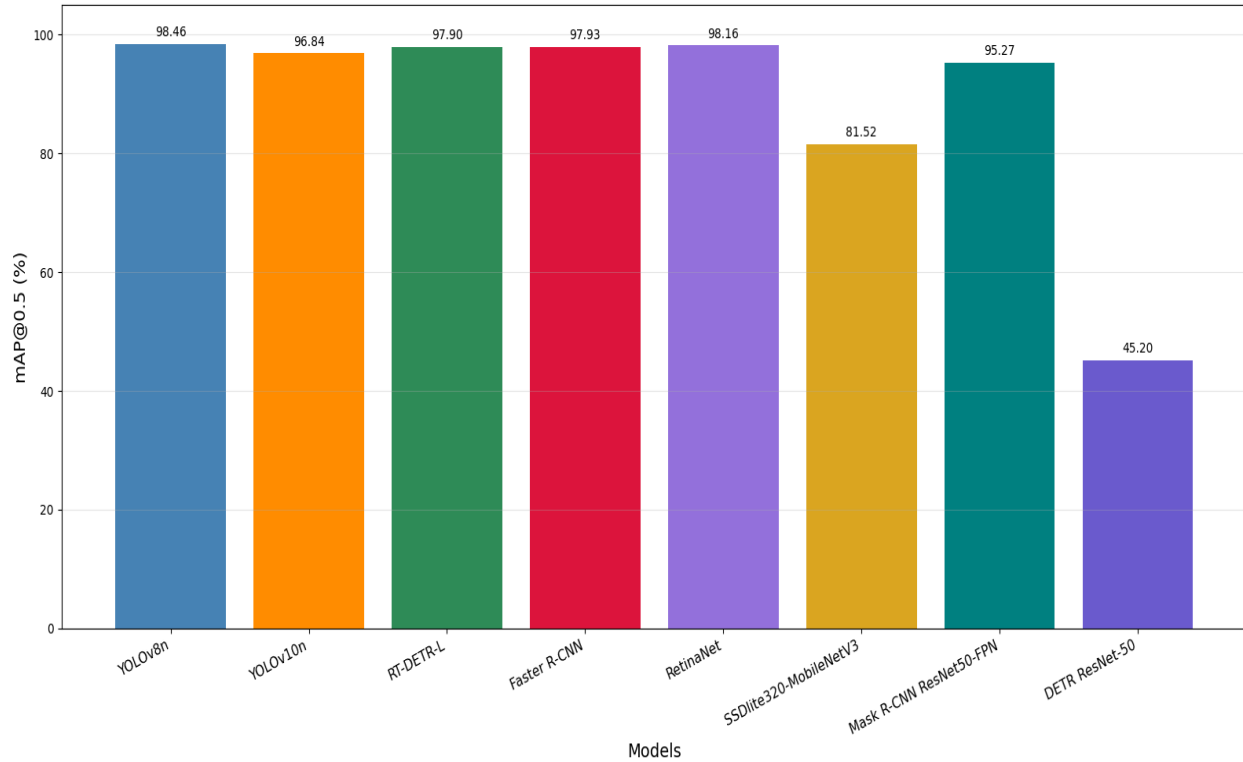


Figure 4. 2: mAP@0.5 Comparison Across Models

The highest mAP@0.5 was achieved by YOLOv8n with 98.46%. This means that YOLOv8n produced the strongest overall detection performance under the standard overlap threshold. The second-best result was obtained by RetinaNet with 98.16%, followed very closely by Faster R-CNN with 97.93% and RT-DETR-L with 97.90%. YOLOv10n also achieved a strong value of 96.84%. Mask R-CNN ResNet50-FPN achieved 95.27%, which is still strong but clearly below the best-performing models. The weakest results were produced by SSDlite320-MobileNetV3 with 81.52% and especially DETR ResNet-50, which achieved only 45.20%.

These findings have clearly shown that the majority of the models were successfully able to learn the underwater detection task effectively, but the success rate varied significantly depending on the models. These four models like YOLOv8n, RetinaNet, Faster R-CNN, and RT-DETR-L performed at a very high level. This suggests that the underwater dataset is enough suitable for training strong detectors which provided that the model architecture is capable enough to handle small-object localization and underwater image degradation. On the other hand, the lower result of SSDlite shows the cost of prioritizing lightweight deployment, while the very weak mAP results

of DETR highlights that this configuration of DETR deep learning model was not suitable for the dataset under the training conditions used in this thesis.

4.3.2 mAP@0.5:0.95

The mAP@0.5:0.95 metric is stricter than mAP@0.5 metric because it calculates average precision for across several IoU thresholds instead of using only 0.5. It is a multi-threshold evaluation, which offers a more in-depth and demanding assessment of object detection performance (Everingham et al., 2010; Padilla et al., 2020). For that reason, this metric provides a more demanding view of localization quality. The model works well here, not only when detecting objects, but also at the level of the more accurate positioning of the boxes.

Under this stricter metric, RT-DETR-L achieved the best result with 82.06% while the YOLOv8n model followed very closely with 81.85%, and YOLOv10n achieved 80.99%. the other models such as RetinaNet achieved 79.26% and Faster R-CNN and Mask R-CNN ResNet50-FPN achieved 74.80% and 74.32% respectively. In contrast, SSDlite320-MobileNetV3 dropped to 47.34%, and DETR ResNet-50 again produced the weakest result at 31.38%.

This is the change in ranking from mAP@0.5 to mAP@0.5:0.95 that is significant. For easier threshold, YOLOv8n was the best 1 and for stricter localization, RT-DETR-L performed best. This indicates that YOLOv8n is the better practical, all-round detector, but RT-DETR-L has a better performance in terms of accuracy for bounding boxes. For the more severe metric, it was again found that RetinaNet was very competitive, reinforcing the belief that a feature pyramid-based one-stage detector, when well trained, is capable of effectively detecting small targets in underwater scenarios.

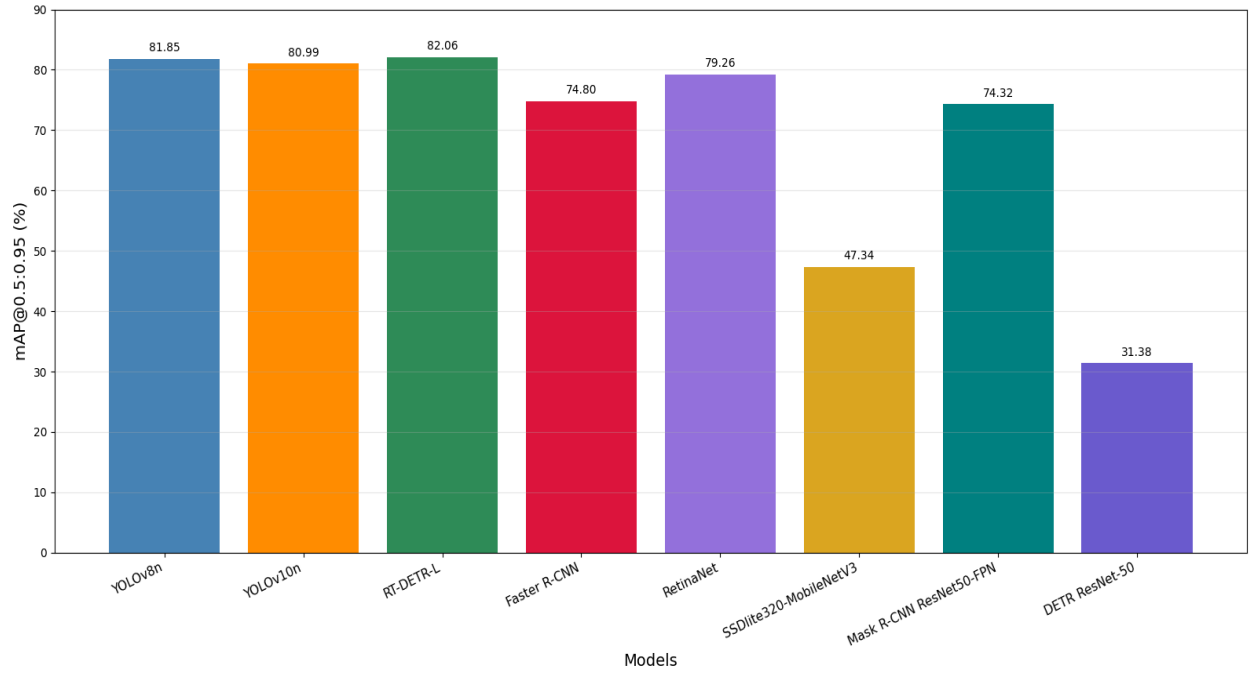


Figure 4. 3: mAP@0.5:0.95 Comparison Across Models

The changing in rank from mAP@0.5 to mAP@0.5:0.95 is important. The metrics results of YOLOv8n model were the strongest model under the easier threshold, but RT-DETR-L became the best model when the localization requirements became stricter. This point suggests that RT-DETR-L model may produce more precise bounding boxes, even though YOLOv8n remains the stronger practical in terms of all-round detectors. While RetinaNet also remained as highly competitive under stricter metric, which supports that the view that feature pyramid-based one-stage detectors can handle small underwater targets effectively when properly trained.

DETR ResNet-50 consistently performed poorly on both measures and the difference was even more evident in the localisation measure. It seems that the actual DETR setup was not well suited for the target data set and task setting.

4.4 Comparison Based on Precision, Recall, and Detection Accuracy

In addition, this thesis not only performs mAP-based evaluation, but also used a custom IoU-based matching logic to compute precision, recall, and detection accuracy consistently across all of the models applied in this study. This is important because all models are judged using the same confidence and IoU thresholds, which makes the comparison more unique and more practical.

4.4.1 Precision

Precision basically measures how many predicted detections were actually correct. In other words, it reflects how well a model can avoid false positives. In underwater scenes with cluttered backgrounds, low contrast, and visually confusing structures, precision is especially important because many false detections can reduce the practical usefulness of the detector.

From the figure 4.3 shows that the highest precision was achieved by YOLOv8n with 96.06%. The other models YOLOv10n and RT-DETR-L were almost identical, with the value of 94.63% and 94.62%, respectively. Another model RetinaNet followed with 92.73%, and Faster R-CNN achieved 91.40%. But the figure clearly indicates that Mask R-CNN ResNet50-FPN had noticeably lower precision result at 80.91%, while SSDlite320-MobileNetV3 achieved 77.23%. The weakest result came from DETR ResNet-50, which achieved only 64.53%.

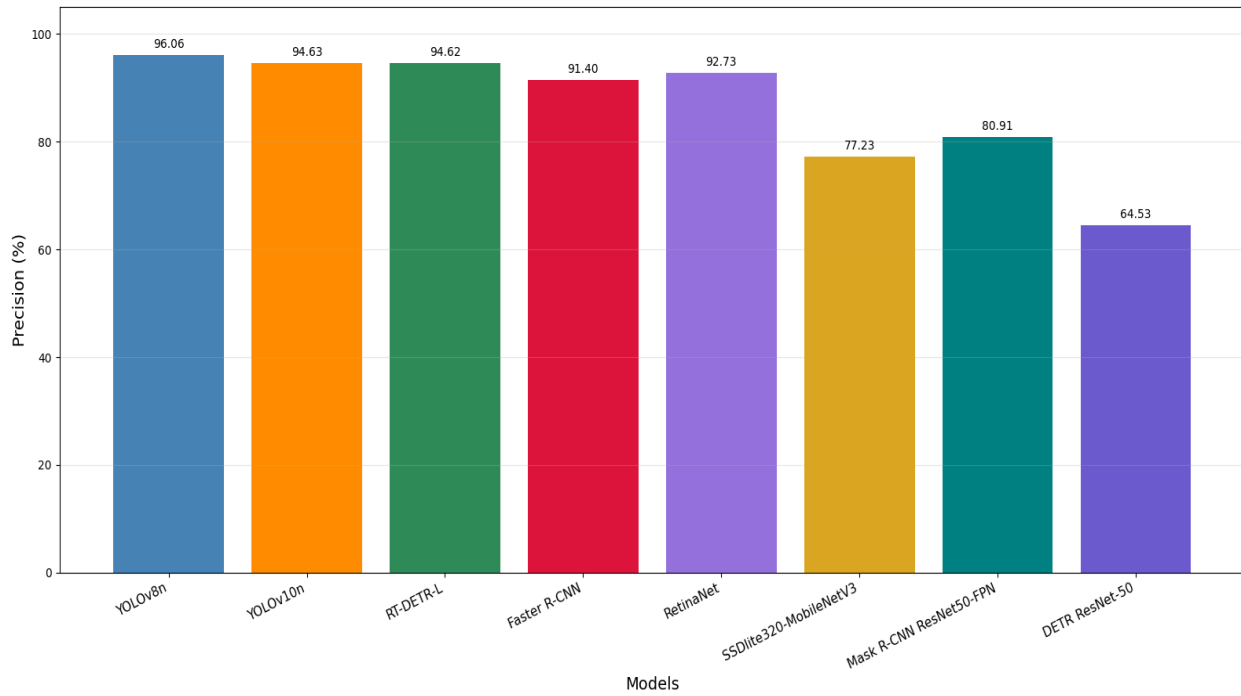


Figure 4. 4: Precision Comparison Across Models

These results highlight that YOLOv8n was the most successfully reliable model in terms of correct positive detections. This means it only generated very few false positive results than the other models. In addition, RT-DETR-L and YOLOv10n were also very strong in this respect. But Mask R-CNN and SSDlite were less reliable, suggesting that these models produce very large number of incorrect detections compared to others. DETR again showed substantial weakness.

4.4.2 Recall

Recall is another important metric which measures how many real objects were successfully detected. In underwater detection, this metric is particularly significant because failing to detect a

small underwater object may be more harmful than producing an occasional false positive. High recall means the model is successful in detecting as many real targets as possible.

In terms of recall metric, the highest recall was achieved by Faster R-CNN with 97.70%. RT-DETR-L followed closely with 97.01%, and RetinaNet achieved 96.78%. Additionally, YOLOv8n also performed strongly with 95.40%, while another model like Mask R-CNN ResNet50-FPN achieved 94.48%. YOLOv10n was somewhat low results at 89.20%, while SSDlite320-MobileNetV3 reached only 67.82%. But DETR ResNet-50 again had the weakest result with 39.31% which indicates that DETR ResNet-50 model is not suitable for this dataset.

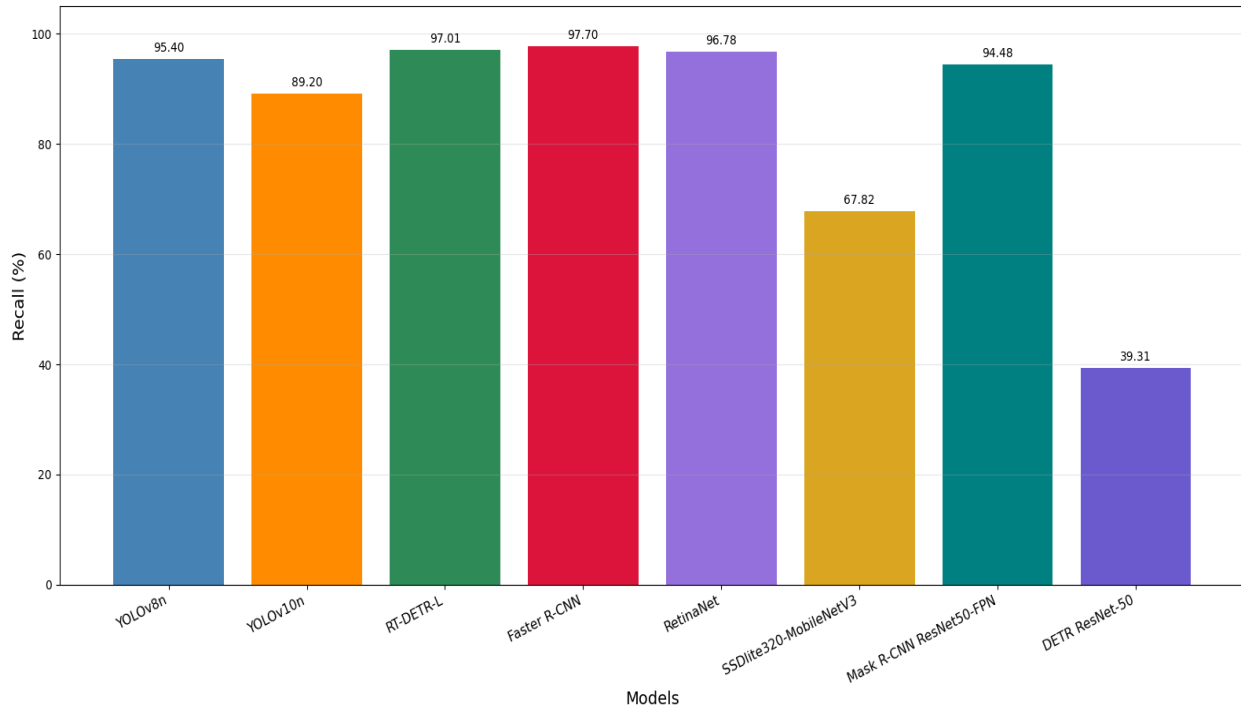


Figure 4. 5: Recall Comparison Across Models

These results expressed that Faster R-CNN was the strongest model in terms of finding existing objects. It failed to detect just fewer targets than the other models. Other model like RT-DETR-L and RetinaNet also showed excellent recall, which supports the view that these models are strong to identifying small underwater targets in complex backgrounds. DETR, by contrast, missed many targeted objects and therefore is not suitable for practical underwater detection under the experimental settings used in this study.

4.4.3 Detection Accuracy

Detection accuracy in this thesis refers to the percentage of filtered predictions that correctly matched a ground-truth object of the same class under the defined IoU threshold. In the notebook

outputs, the ranking of detection accuracy was identical to the ranking of precision, because the evaluation logic counted correct predictions in the same filtered prediction space.

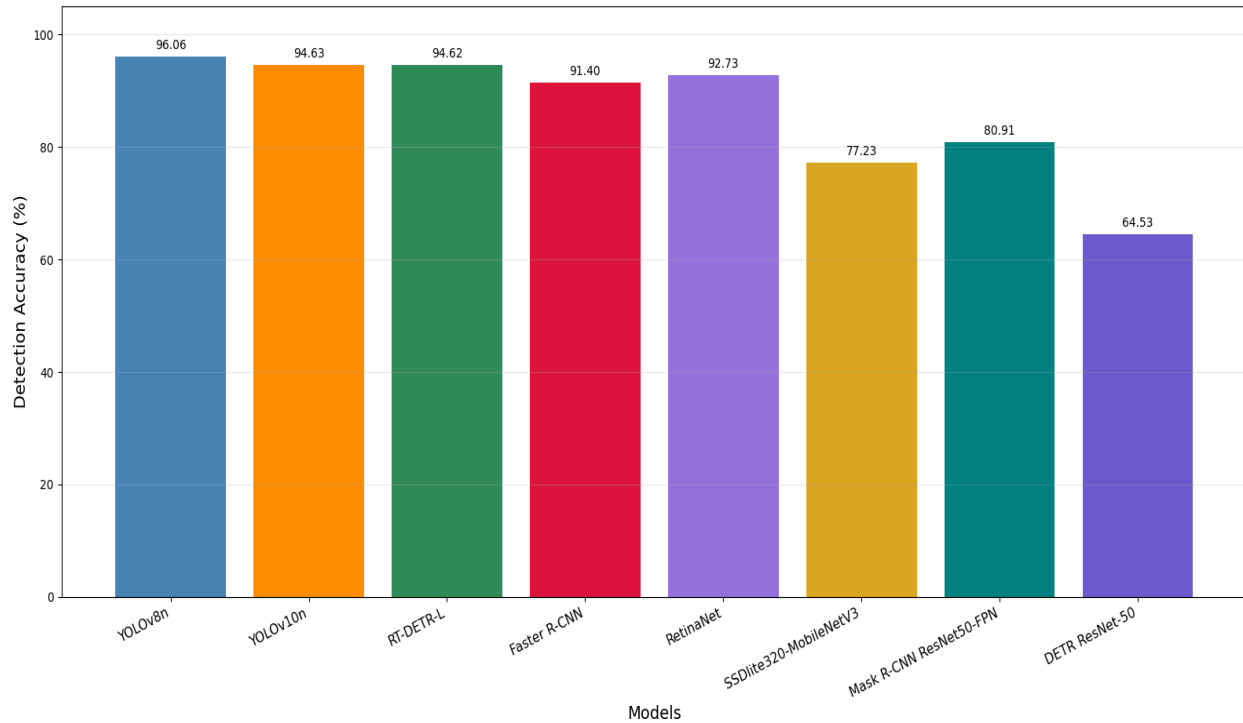


Figure 4. 6: Detection Accuracy Comparison Across Models

The highest detection accuracy was achieved by YOLOv8n with 96.06%. This was followed by YOLOv10n with 94.63%, RT-DETR-L with 94.62%, RetinaNet with 92.73%, and Faster R-CNN with 91.40%. Mask R-CNN ResNet50-FPN reached 80.91%, SSDlite320-MobileNetV3 achieved 77.23%, and DETR ResNet-50 again produced the weakest result with 64.53%.

Moreover, precision, recall, and detection accuracy show that the models differ in behavior while YOLOv8n is the most balanced model, combining very high precision and strong recall. Faster R-CNN is the strongest recall-oriented model, meaning it detects the greatest share of existing objects, but at some cost in precision and runtime. RT-DETR-L is highly competitive across all three measures and appears to be one of the most robust models overall.

4.5 Comparison Based on Computational Efficiency

4.5.1 Frames Per Second and Average Inference Time

In real underwater systems, speed is often just as important as accuracy. A detector that is very accurate may still be unsuitable for deployment if it is too slow to support inspection or robotic navigation tasks.

YOLOv8n with the inference time of 0.0088 seconds and the FPS of 113.67 seconds was the fastest one in this study. The other models also performed well. For example, SSDlite320-MobileNetV3 ran in 0.0197 seconds and 101.42 FPS, while YOLOv10n ran in 0.0101 seconds and 98.80 FPS. Although slower at 0.0416 seconds and 24.05 FPS, RT-DETR-L was also considerably faster than the more robust torchvision based detectors. DETR ResNet-50 performed best with 16.21 FPS, RetinaNet performed with 14.11 FPS, Mask R-CNN ResNet50-FPN performed with 11.93 FPS, and Faster R-CNN was the slowest with 11.75 FPS.

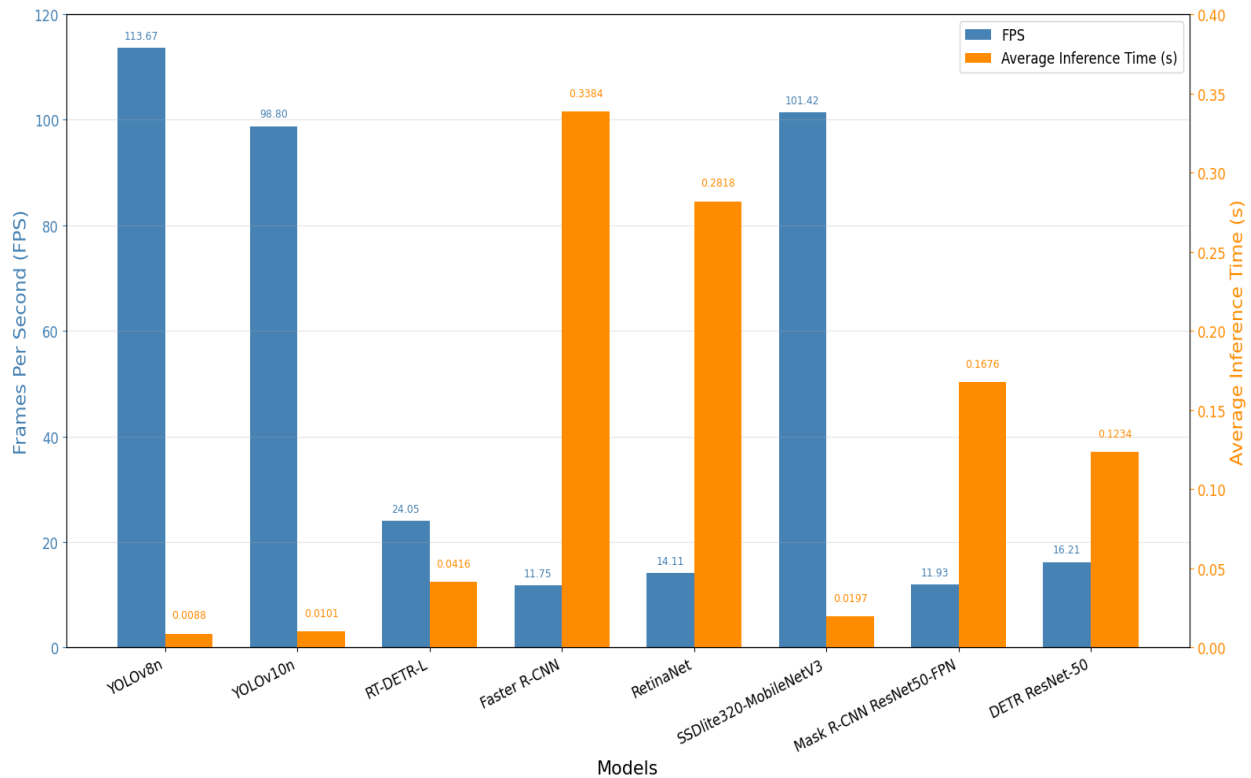


Figure 4. 7: FPS and Average Inference Time Comparison Across Models

From these results, it is very clear that the YOLO models are lightweight compared to two-stage models. YOLOv8n, YOLOv10n, and SSDlite are very suitable for fast deployment. The intermediate position is the RT-DETR-L, which is much faster than the YOLO models, but much slower than Faster R-CNN and Mask R-CNN. DETR ResNet-50 is faster than Faster R-CNN, however, due to its poor detection accuracy, this advantage is not enough to compensate.

4.5.2 Parameter Count

Parameter count is an important indicator of model complexity and deployment difficulty. Generally, the larger the model, the more memory and computing power it will need, and the smaller the model, the more easily it can be deployed on small hardware.

From the figure 4.7 shows that the smallest model was YOLOv10n achieved with 2,775,520 parameters, and other small models such as YOLOv8n achieved with 3,157,200 parameters and SSDlite320-MobileNetV3 with 3,792,380 parameters. At the other end, Mask R-CNN ResNet50-FPN which had 43,949,305 parameters, DETR ResNet-50 had 41,502,923 parameters, and Faster R-CNN had 41,324,786 parameters. RT-DETR-L and RetinaNet were also relatively heavy model, with 32,970,476 and 32,272,419 parameters, respectively.

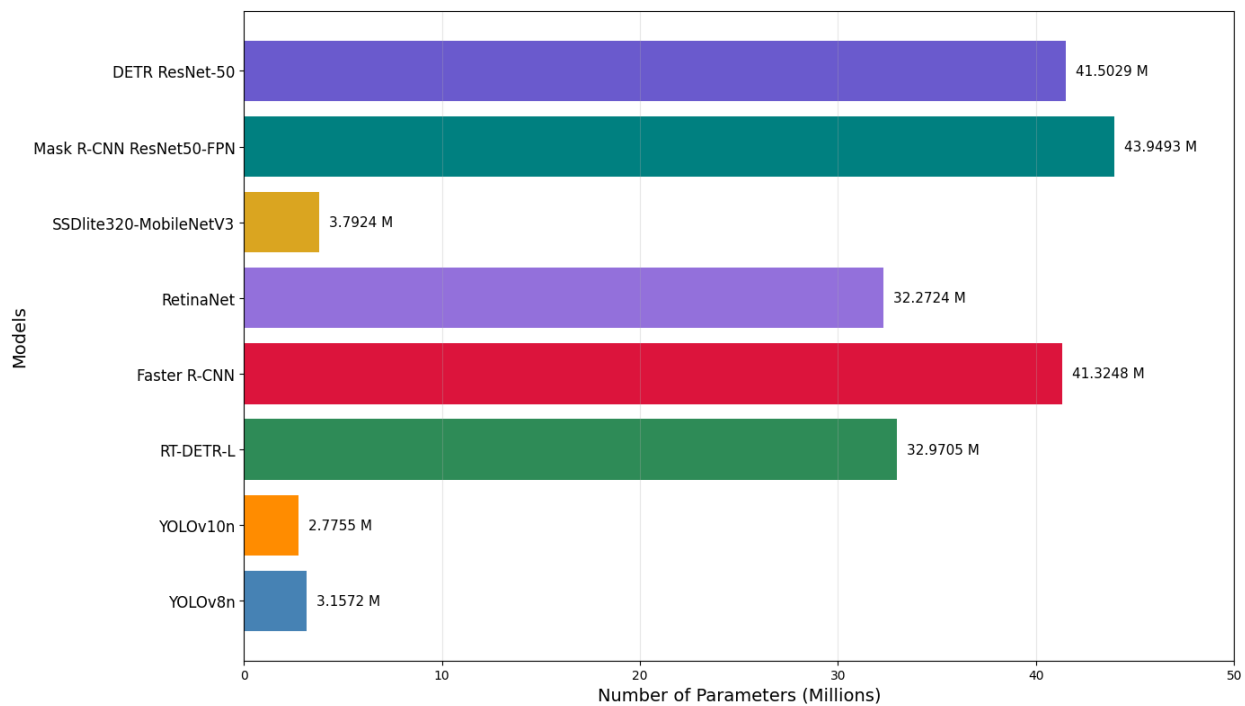


Figure 4. 8: Parameter Count Comparison Across Models

This is one of the reasons the YOLO models are so efficient because they have few parameters and great detection performance. Mask R-CNN, Faster R-CNN, and DETR are considerably heavier models that are not ideal for deployments that have limits on computing power.

4.6 Comparison Based on Training Time

Training time is another important metric of practical comparison in this study. Although training is usually done offline, it indicates that a model that requires much longer training still imposes a heavier research and deployment burden.

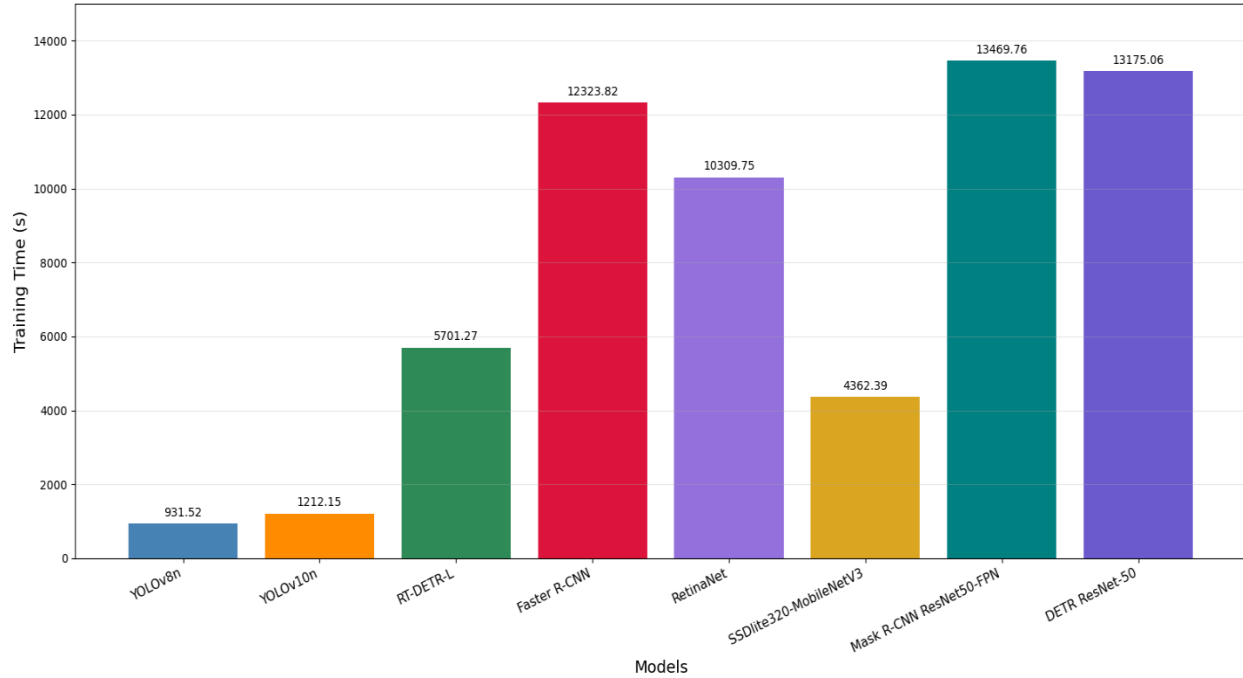


Figure 4. 9: Training Time Comparison Across Models

In the figure 4.8 shows that the shortest training time was achieved by YOLOv8n, which completed training just in 931.52 seconds. YOLOv10n was second with 1212.15 seconds. The time for SSDlite320-MobileNetV3 was 4362.39 seconds and RT-DETR-L took 5701.27 seconds. The longest training time was for RetinaNet was 10309.75 seconds, Faster R-CNN model training completed in 12323.82 seconds, and DETR ResNet-50 finished in 13175.06 seconds and Mask R-CNN ResNet50-FPN model training completed in 13469.76 seconds.

These results align with the comparisons between parameters and inferences. The faster models will also train faster, and the heavier models will take significantly longer to train. YOLOv8n and YOLOv10n are especially promising for research workflows and quick experimentation.

4.7 Accuracy-Speed Trade-off

One of the most important goals of this thesis is to understand the trade-off between accuracy and computational speed efficiency. This trade-off is most clearly seen by comparing FPS and mAP@0.5 directly.

YOLOv8n has the best performance overall as seen in the accuracy-speed scatter plot. This model archived the most mAP@0.5 and FPS. YOLOv10n is also very practical, and very fast and accurate. For strict localization RT-DETR-L is in a different stage because it is the slowest model in FPS, but it is the best at localization and the overall detection quality is very high. RetinaNet and Faster R-CNN fall on the high accuracy but low speed end of the spectrum. Mask R-CNN

ResNet50-FPN has similar structure but with less accuracy. In terms of speed, SSDlite320-MobileNetV3 is undoubtedly less accurate. DETR ResNet-50 is in the lowest spot in terms of accuracy and an average speed.

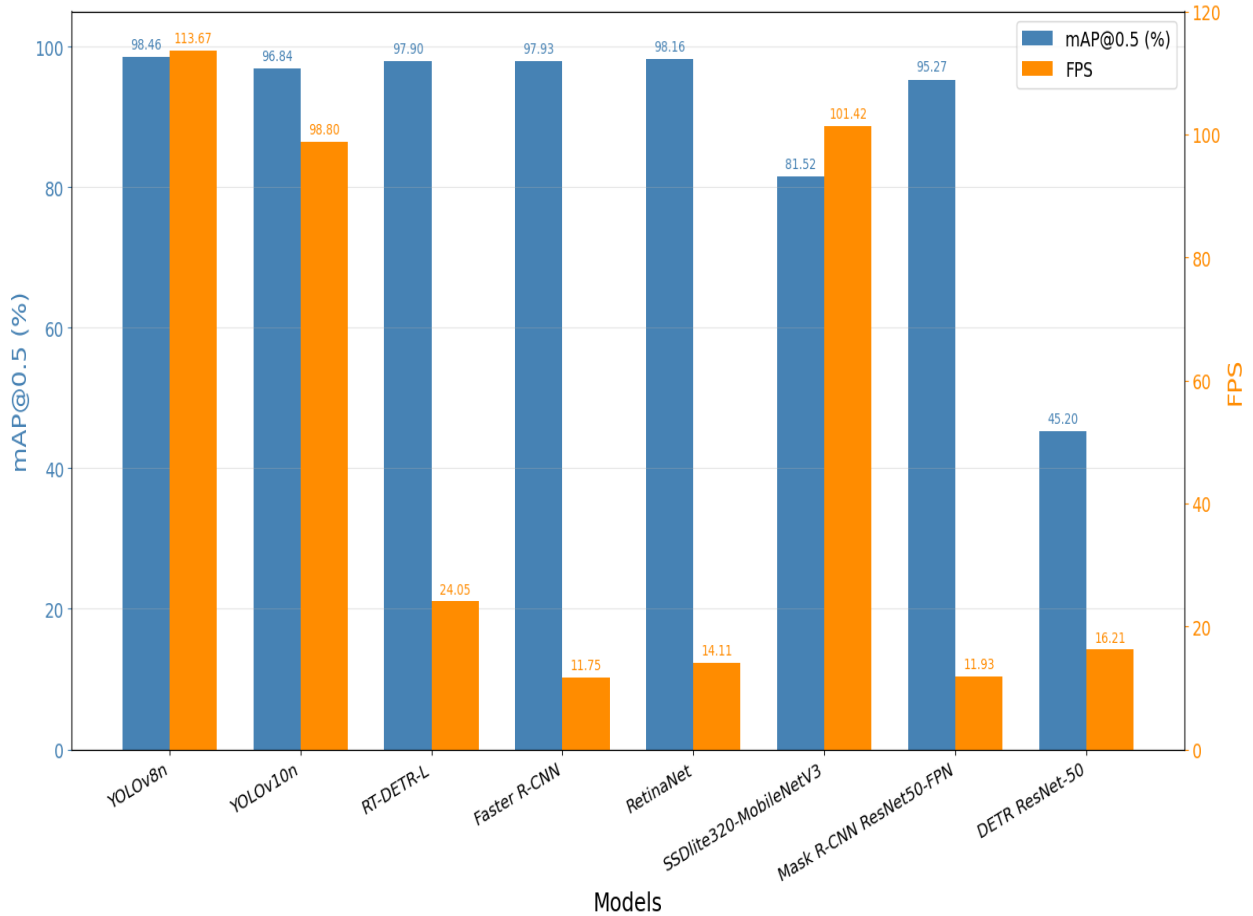


Figure 4. 10: Accuracy-Speed Trade-off

This figure 4.9 highlights the main experimental conclusion is very clear. Basically, model selection depends on the intended application. If real-time deployment is the main priority, then YOLOv8n is the best choice. If stricter localization quality is more important and lower FPS can be accepted, the RT-DETR-L becomes highly competitive. If the high degree of recall is most important, then Faster R-CNN is also a good choice, although its speed penalty is large.

4.8 Comparative Interpretation of the Results

The results enable the models to be put into easily recognizable practical categories. The first group are YOLOv8n and YOLOv10n. The models are the best practical detectors in the study due to their high detection quality and very high speed. Among them, YOLOv8n is the best model overall. It outperforms other methods on the highest mAP@0.5, the precision, the detection accuracy, the fastest inference time, the highest FPS and the shortest training time. YOLOv10n also did great but slightly less well than YOLOv8n in Recall and strict mAP.

The second group is represented by RT-DETR-L. This model was the best model in strict localization quality based on the $mAP@0.5:0.95$. It also achieved very high recall and very strong overall accuracy. However, it is much more computationally intensive than the YOLO models both in training and inference. Hence, RT-DETR-L can be regarded as a high-quality detector for the case where further localization accuracy is required and some reduction in speed is acceptable.

The third class are the heavier classical detectors, namely Faster R-CNN, RetinaNet and Mask R-CNN ResNet50-FPN. The results of these models confirm that classical architectures provide strong detection quality, but are less efficient than the best one-stage detectors. Faster R-CNN had the best recall, meaning that it was best at identifying real objects. The $mAP@0.5$ and $mAP@0.5:0.95$ of RetinaNet are very competitive, which makes it a very accurate one-stage model based on accuracy. It could get good detection quality, but its precision was lesser than Faster R-CNN and RetinaNet and its training time was the longest among all the models.

The fourth group comprises of SSDlite320-MobileNetV3 and DETR ResNet-50, but the two models performed very differently. SSDlite was very fast and light, but with a much lower detection accuracy. It is only useful for very limited deployment scenarios. In contrast, DETR was not accurate, and not particularly efficient of the options available. In the experimental setting employed in this thesis, the DETR ResNet-50 model was the least powerful of all models.

4.9 Best Model Depending on Application Goal

Based on the results, the optimum model is dependent on the application scenario. In summary, YOLOv8n is the best model in this study if the ultimate target is to have high detection accuracy, high speed, and low computation cost. It offers the best overall detection quality and the fastest inference speed and most practical deployment efficiency. RT-DETR-L, on the other hand, would be the optimal choice when the focus is on the localization precision of the bounding boxes instead of speed, as it outperformed the other compared models on $mAP@0.5:0.95$.

However, if the primary objective is to identify as many objects as possible that exist in the real world, the best model to consider is Faster R-CNN as it has the highest recall. Moreover, RetinaNet is also a suitable choice if the objective is the high level of detection accuracy with a classical one-stage detector, although it is slow when compared with the YOLO based models. However, for very limited hardware, SSDlite320-MobileNetV3 might be useful as it is small and quite fast, but there is a loss of detection accuracy. Therefore, when both practical speed and strong underwater detection performance are required together, YOLOv8n remains the most suitable overall model in this study.

4.10 Chapter Summary

In this chapter, the comparative results of eight underwater object detection models evaluated under a common framework. The models varied greatly in their ability to detect and computational efficiency. The best overall model was YOLOv8n with all the best parameters in mAP , precision,

detection accuracy, speed and training time. RT-DETR-L performed best in strict localization, and Faster R-CNN performed best in terms of recall. RetinaNet had a strong and competitive performance as well, particularly in the mAP metrics. Although SSDlite320-MobileNetV3 was helpful as a light and fast deployment oriented model, DETR ResNet-50 performed the worst overall in the setting of this study.

Overall, the results support the central argument of the thesis because there is no single best detector model for every requirement, and model selection depends on the balance between accuracy, localization quality, speed, and computational cost. However, for this dataset and settings which were used in this work, YOLOv8n provides the strongest overall solution for detecting small underwater objects in deep-sea environments.

Chapter 5 Conclusion and Future Work

5.1 Conclusion

This thesis compared the performance of eight deep learning-based object detection models for underwater small object detection. The evaluated models were YOLOv8n, YOLOv10n, RT-DETR-L, Faster R-CNN, RetinaNet, SSDlite320-MobileNetV3, Mask R-CNN ResNet50-FPN and DETR ResNet-50. The main goal of the study was to explore how well and quickly the current state-of-the-art object detection architectures work in underwater environments with multiple small and visually complex objects.

Low contrast, low illumination, color distortion, suspended particles and occlusion are the main problems that are of challenge in underwater object detection. Light attenuation and scattering phenomena are major environmental factors that significantly lower the image quality in underwater environments (Jaffe, 2002; Jian et al., 2024). These environmental conditions greatly influence the feature extraction and object localization performance. Furthermore, small objects under water occupy small areas in the pixels which makes the detection of such objects more challenging than conventional object detection. Thus, it is critical to choose a proper detection architecture to ensure an effective underwater monitoring and inspection system.

The experimental results demonstrated that lightweight YOLO-based detectors achieved the best balance between detection accuracy and computational efficiency. Among all evaluated models, YOLOv8n achieved the highest overall performance with an mAP@0.5 score of 98.46%, precision of 96.06%, recall of 95.40%, and FPS of 113.67. The model also maintained the shortest inference time while using a relatively small number of parameters. These results indicate that YOLOv8n is highly suitable for real-time underwater object detection applications where both speed and accuracy are important.

YOLOv10n also demonstrated strong performance with competitive detection accuracy and high processing speed. RT-DETR-L achieved high detection accuracy as well, especially in terms of recall performance, but required significantly higher computational resources and longer training time due to its transformer-based architecture.

At the detection stage, Faster R-CNN and Mask R-CNN ResNet50-FPN led to high detection accuracy and stable localization results. But both models needed significantly more training and inferring time than lightweight one-stage detectors. Overall the accuracy of RetinaNet was high and the performance was well balanced, but it was still a bit more computationally intensive than the YOLO-based ones.

For edge-device deployment and resource-constrained systems, SSDlite320-MobileNetV3 proved to be fast with low computational complexity. However, its detection accuracy was lower than most other evaluated models, especially for difficult small-object scenarios. The lowest accuracy for the detection was obtained with DETR ResNet-50 which consumed a large number of

computational resources. This shows that transformer models achieve good performance on complex environments, which is likely to require larger data sets and longer optimization time (Carion et al., 2020; Zhao et al., 2024).

The comparative analysis also revealed that there is no one best detector for all underwater applications. On the contrary, the choice of a detection model depends on the application needs, hardware provided and real-time processing requirements. Depending on the application, lightweight models like YOLO are better suited for real-time applications, whereas more complex models like Mask R-CNN or RT-DETR-L are better suited for applications that need higher localization accuracy and representativeness of features.

Overall, this thesis provides a practical benchmark comparison of existing deep learning-based underwater object detection models with a common experimental setting. These results can serve as helpful guidelines for future underwater computer vision studies and in real-world applications such as marine monitoring, underwater robotics, undersea environmental scanning, and autonomous underwater navigation.

5.2 Limitations of the Study

Although the thesis achieved its primary objectives as well as several limitations should be acknowledged. First, the study was conducted using a single underwater dataset containing six object classes. The dataset provided was realistic underwater conditions but might not cover all the underwater environments that exist in real-world applications. The performance of the model may vary depending on ocean conditions, lighting variations, turbidity and the type of object.

Second, the same training configuration and hyperparameter setting were used for all experiments. The models were not extensively optimized by using advanced hyperparameter tuning, neural architecture search, or large-scale parameter optimization. So, there are some models that might perform better with a better setup.

Third, computational experiments had to rely on the availability of the GPUs and the amount of training time allowed. The number of 30 epochs was found to be sufficient for convergence for most of the models, but some models like DETR and transformer based detectors may require longer training times and larger batch sizes.

Fourth, the study focused only on bounding-box object detection performance. Although Mask R-CNN supports instance segmentation, segmentation masks were not evaluated in this thesis. The research did not cover tasks like semantic segmentation, depth estimation, object tracking, and multi-object trajectory analysis.

Fifth, the evaluation was conducted mainly in terms of quantitative measures like mAP, precision, recall, FPS, and inference time. There were only limited qualitative analysis and robustness testing on human with severe degradation underwater.

Lastly, the experiments were not performed with actual underwater robotic systems. Hence, further real-time deployment issues including hardware latency, energy consumption, communication delay and integration of underwater sensors were not thoroughly explored.

5.3 Future Work

To enhance the performance of underwater object detection and future practical deployment, multiple potential research directions can be followed to explore.

First, future research may investigate larger and more diverse underwater datasets containing additional object categories, environmental conditions, and imaging scenarios. More diverse datasets could improve model generalization and robustness across different underwater applications.

Second, more sophisticated image augmentation and underwater image enhancement methods can be incorporated in the training path. Color correction, dehazing, contrast enhancement, synthetic data generation and generative adversarial networks (GANs) can enhance the visibility and enable accurate feature detection in poor underwater conditions.

Third, newer transformer-based detection models and hybrid CNN-transformer models can be studied in the future. In recent times, vision transformers have shown great performance in learning global contextual features for complex visual tasks (Dosovitskiy et al., 2021).

Fifth, the work could be further extended to instance segmentation, object tracking, and video based underwater scene understanding in the future. The incorporation of temporal tracking and object detection could enhance autonomous underwater navigation systems and long-term marine monitoring systems.

Fifth, future research may extend the work toward instance segmentation, object tracking, and video-based underwater scene understanding. Combining object detection with temporal tracking could improve autonomous underwater navigation and long-term marine monitoring systems.

Sixth, lightweight model compression and optimization techniques such as quantization, pruning, and knowledge distillation may be explored for deployment on embedded underwater hardware and edge devices with limited computational resources.

Finally, future work may focus on real-world deployment and field testing using autonomous underwater vehicles (AUVs), remotely operated vehicles (ROVs), and underwater robotic platforms. Real-time underwater deployment experiments would provide a more realistic evaluation of model robustness, latency, and operational reliability in practical marine environments.

In conclusion, the detection of underwater objects is a current and ongoing research topic. These advancements will continue to improve in the future, thanks to ongoing advancements in intelligent underwater vision systems. Further developments in deep learning architectures,

underwater imaging techniques, and computational hardware are likely to further increase the performance and utility of intelligent underwater vision systems in the future.

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